

Evaluation of two methods to reduce legume-related flatulence through enzymatic digestion of flatulence factors

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Shraddha Ranganathan
Matriculation number: 22553

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Supervised by
Frau Prof. Dr. rer. nat. habil. Mònica Palmada Fenés

Co-Supervised by
Lucas Evers

Abstract

Legumes are nutritionally equivalent to many meat products, and can be used to supplement or replace meat in daily diets. This is beneficial for a variety of reasons, such as: i) growing legumes can help to reduce acidification of soil, global warming potential and energy use; ii) livestock is taxing on the environment in terms of adding to the volume of greenhouse gases and nitrification of soil while legume crops fix soil nitrogen.

Consumers can be opposed to adding legumes to their diet due to the perception of legumes causing flatulence. Intestinal gas buildup, bloating, cramps, abdominal pain and flatulence are caused by raffinose family oligosaccharides (RFOs) which cannot be digested in monogastric organisms such as humans. They are therefore broken down by microflora in the intestine; this bacterial digestion releases large volumes of hydrogen, which causes flatulence.

The human body lacks the enzyme required to break down these RFOs — α -galactosidase. This experiment evaluated two methods of applying α -galactosidase to RFOs before they reach the intestinal microflora. The first method evaluates the effectiveness of enzymatically digesting the legumes before consumption. The second evaluates the effectiveness of the enzyme supplement Beano, which applies the enzyme to RFOs in the stomach.

Experimental data showed that enzymatically digesting raw flours does significantly reduce the amount of RFOs in the legume. Similarly, Beano also reduces RFOs significantly. In 4 out of the 6 legumes sampled, there was no significant difference between the two methods.

In order to consider the methods for commercial use, other factors (such as economic, logistical, etc.) must also be considered. At the outset, it appears that taking an enzyme supplement such as Beano might be more economically viable in the long term for the

consumer, since processing costs for the flatulence free legumes would drive up the price of (normally cheap) legumes.

There is an increase in the amount of people who are giving up meat for environmental and other reasons. For these consumers, as well as those who come from cultures that integrate legumes in their cuisine, the removal of flatulence factors from this nutrition-rich food group will be very beneficial.

Contents

1. Introduction	1
1.1 Premise: Environmental problems in the EU, especially the Netherlands	1
1.2 High emission industry: meat industry	1
1.3 Legumes as a means of reducing meat consumption	2
1.4 Relationship between legume consumption and flatulence	5
1.5 Action of α-galactosidase on raffinose family oligosaccharides	7
1.6 Possible solutions to flatulence caused by legume-consumption	10
1.7 Enzymatic methods of digesting flatulence factors	11
1.8 Structure of this work	12
2. Aim of the work	13
3. Materials and Methods	14
3.1 Pre-digestion of flatulence factors	17
3.1.1. Materials used	17
3.1.2 Protocol	18
3.2 Digestion of flatulence factors in a simulated stomach environment	18
3.2.1 Materials used	18
3.2.2 Protocol for constructing the simulated stomach environment	19
3.2.3 Digestion of flatulence factors in the simulated stomach environment	19
3.3 Assay and Measurement of flatulence factors	20
3.3.1 Materials used	21
3.3.2 Protocol	22
3.3.3 Calculation of flatulence factors	22
3.4 Statistical analysis	25
4. Results	25

4.1 Effectiveness of pre-digestion across various types of legumes	27
4.2 Extent of RFOs reduction in a stomach environment by using Beano	28
4.3 Comparison between two methods	31
5. Discussion	33
5.1. Effectiveness of pre-digestion method on different legumes	33
5.2. Extent of reduction of RFOs by using Beano	34
5.3. Comparison of two methods	37
5.4. Non-enzymatic methods of reducing RFOs in legumes	39
5.5. Flatulence-free legumes in an environmental context	41
6. Outlook	42
7. References	45
8. Appendix	53
8.1 Specifics about Beano	53
8.2 Specifics regarding other enzyme supplements	56
8.3. “No gas beans”	58
9. Statutory declaration	60

1. Introduction

1.1 Premise: Environmental problems in the EU, especially the Netherlands

Western Europe has a nitrogen problem: over 90% of vulnerable ecosystems receive more than the critical load of nitrogen (Steinfeld, 2006). The Netherlands is particularly affected, since it has now reached a nitrogen crisis. The Dutch government has responded to this by proposing two *Spoedwetten* (emergency laws): the first mandates that the speed limit on freeways be lowered from 110 km/h to 100 km/h during the day; the second suspends permits for construction projects that pollute the atmosphere with nitrogen compounds and harm nature reserves (Tweede Kamer der Staten-Generaal, 2019). These laws came into effect in December, 2019.

This, in combination with the Netherlands' undertaking to reduce CO₂ emissions by 50%, by the year 2030, presents the challenge of approaching environmental problems in a diverse and varied manner. Short term measures to reduce CO₂ emissions include 'greening' the tax system and providing more offshore space for wind energy resources. (Statistics Netherlands (CBS), 2018)

This demonstrates the need for both conventional and unconventional approaches to tackling environmental problems. In this case, it is important to evaluate industries which i) have high emissions, and ii) do pollute their immediate surroundings.

1.2 High emission industry: meat industry

The meat industry, for instance, satisfies both requirements. By nature of their biology, ruminants (such as cows, sheep, and goats) do produce significant volumes of methane. Such domestic ruminants are responsible for 25% of the emissions linked to human activities. Additionally, the urine and manure from livestock contains nitrogen, which leaches other nutrients out of the soil (Makkar and Vercoe, 2007; Steinfeld, 2006). Furthermore, agricultural run-off containing nitrogen can contribute to the eutrophication

of nearby water bodies (Khan and Mohammed, 2014). This, in turn, affects water quality, soil quality, as well as the health of local flora and fauna.

This study keeps the Dutch context in mind while discussing the results. However, the issues related to the meat industry are global. For instance, total global meat production increased between 1980 and 2007, from 136 to about 285 million tons (Den Hartog and Sijtsma, 2011). Considering direct (methane emissions from ruminant gastric functions, nitrogen emissions from urine and manure) and indirect (packaging of meat products, transport, etc.) factors, this implies that this 27-year period has seen massive amounts of environmental impacts due to the increase in demand for meat products.

One way to reduce the environmental impact of the meat industry would be for meat consumption to be reduced; reducing the demand for meat, and thus reducing pollutants and emissions caused by producing, packaging, and transporting meat products.

1.3 Legumes as a means of reducing meat consumption

Meat products are rich in proteins (see Table 1.3.1). The daily protein requirement for a healthy adult is 0.65 grams of good quality protein, per kilo of body weight, per day (Rand et al., 2003). Thus, a healthy adult, weighing 70 kilograms, would require ($0.65\text{ g} \times 70\text{ kg} =$) 45.5 grams of protein per day.

Table 1.3.1: Protein content per gram of food-item (meats).

Source	Protein content (g/g)	Source
Beef steak	0.23	(FDC, 2019)
Pork, Leg Cap Steak	0.21	(FDC, 2019)
Chicken breast	0.20	(FDC, 2019)
Lamb, loin chop	0.24	(FDC, 2019)

If this protein requirement were to be satisfied only by consuming a meat product, this individual would have to eat ($\frac{45.5 \text{ g}}{0.22 \text{ g}} =$) 206.82 grams of a meat product. In this calculation, the figure 0.22 g is used. This is the average amount of protein per gram of meat, as seen from the data in Table 1.1.

Thus, completely removing meat from one's diet without substituting an equally protein-rich food, would be detrimental to one's health.

An equally protein-rich, meatless option is legumes. This category of plants is very diverse: it ranges from peas (*Pisum sativum*) to beans (such as kidney beans, *Phaseolus vulgaris*) to lentils (*Lens culinaris*).

On average, legumes contain 0.24 grams of protein per gram of legume (See Table 1.3.2 - soybeans are excluded as an outlier). As above, assuming a healthy adult weighing 70 kilos was to fulfill their entire protein intake requirement using only legumes, they would need to consume ($\frac{45.5 \text{ g}}{0.24 \text{ g}} =$) 189.58 grams of legumes.

Table 1.3.2: Protein content per gram of food-item (legumes).

Source	Scientific nomenclature	Protein content (g/g)	Source
Beans, Dry, Dark Red Kidney (0% moisture)	<i>Phaseolus vulgaris</i>	0.26	(FDC, 2019)
Chickpeas (garbanzo, bengal gram), mature seeds, raw	<i>Cicer arietinum</i>	0.20	(FDC, 2019)
Peas, green, split, mature seeds, raw	<i>Pisum sativum</i>	0.23	(FDC, 2019)
Red lentils	<i>Lens culinaris</i>	0.25	(FDC, 2019)
Beans, dry, pinto (0% moisture)	<i>Phaseolus vulgaris pinto</i>	0.24	(FDC, 2019)

Mung beans, mature seeds, raw	<i>Vigna radiata</i>	0.24	(FDC, 2019)
Soybeans, mature seeds, dry roasted	<i>Glycine max</i>	0.43	(FDC, 2019)

Thus, it is seen that the daily protein requirement for human adults can be easily satisfied by consuming legumes in place of meat.

However, there exist certain preconceptions about legumes that cause people to reject them from their diets. One example is the paleolithic diet, which has been growing in popularity (Manheimer et al., 2015). This diet “emphasizes an increased consumption of lean meat, fish, shellfish, fruit, vegetables, eggs, nuts, and seeds while excluding grains, legumes, cereals, dairy, processed foods, refined sugars and added salt”, according to Nutrition and Health Info Sheet from UC Davis’ Department of Nutrition (Berggren et al., 2018).

Reasons to avoid legumes, from the Paleo diet perspective, include the presence of phytic acid. They claim that phytic acid binds to nutrients, preventing them from being absorbed. It is also recommended to avoid legumes since they contain lectins, which may be potentially toxic to some consumers (Paleo leap LLC, 2019).

However, these claims are made on a lifestyle blog, without the citation of published research. Other health, wellness and fitness blogs do cite research that provides evidence that phytic acid and lectins may not be detrimental for health, such as the blog “Breaking Muscle” (Taraday, 2018).

Nutritional factors aside, meat is important to the cultural food of many regions. Notably, American policymakers pushed for the rebranding of meat as ‘fuel for soldiers’ during World War II. This propagated the association of meat with masculinity, which propelled sales. Additionally, since meat was rationed for soldiers, the supply for civilians was limited. This also gave meat its ‘elite-food’ status (Chiles and Fitzgerald, 2017).

Legumes (or more commonly, beans), on the other hand, have a history of being seen as a “poor man’s meat”, as described by the author in the book *Beans: A History* (Albala, 2007). The author observes this across many cuisines, stating “In any culture where a proportion of people can obtain protein from animal sources, beans will be reviled as a food fit only for peasants”. Therefore, as a matter of social status, some consumers might be reluctant to (re-)introduce legumes into their diet after having made the switch to meats, which are more expensive and have historically shown higher social class.

There is also a perception of increased flatulence related to the consumption of legumes (Winham and Hutchins, 2011). Individuals wishing to avoid this unpleasant side effect, may eliminate legumes from their diet altogether. Seeing the near-equivalence of the protein quantity in legumes and meat, alongside the other reasons mentioned previously, consumers may choose meat over legumes.

However, in recent times, dietary habits such as veganism and “Meatless Monday” have also gained popularity (Chiles and Fitzgerald, 2017). Certain ethnic cuisines are also being adopted globally, which use legumes in their foods. Some examples are Indian, Brazilian and Persian cuisines (Larijani et al., 2016; Patil et al., 2009). These trends would indicate a future of increased legume consumption. Therefore, individuals making the switch to veganism, or switching out meat for legumes, may also notice an increase in gastric discomfort and flatulence. If this is a chief concern, then there are options available for the consumer who would like to make the switch from meat to beans, while avoiding flatulence.

1.4 Relationship between legume consumption and flatulence

Legumes contain α -galactosides; these are non-reducing sugars with low molecular weights. The simplest of these is raffinose, and the group of similar sugars is called raffinose-family oligosaccharides (RFOs) (Dey, 1980). Other members of this family, also present in legumes, include stachyose, verbascose, ajugose, and ciceritol (Martínez-Villaluenga et al., 2006). Figures 1.4.1 demonstrates the examples.

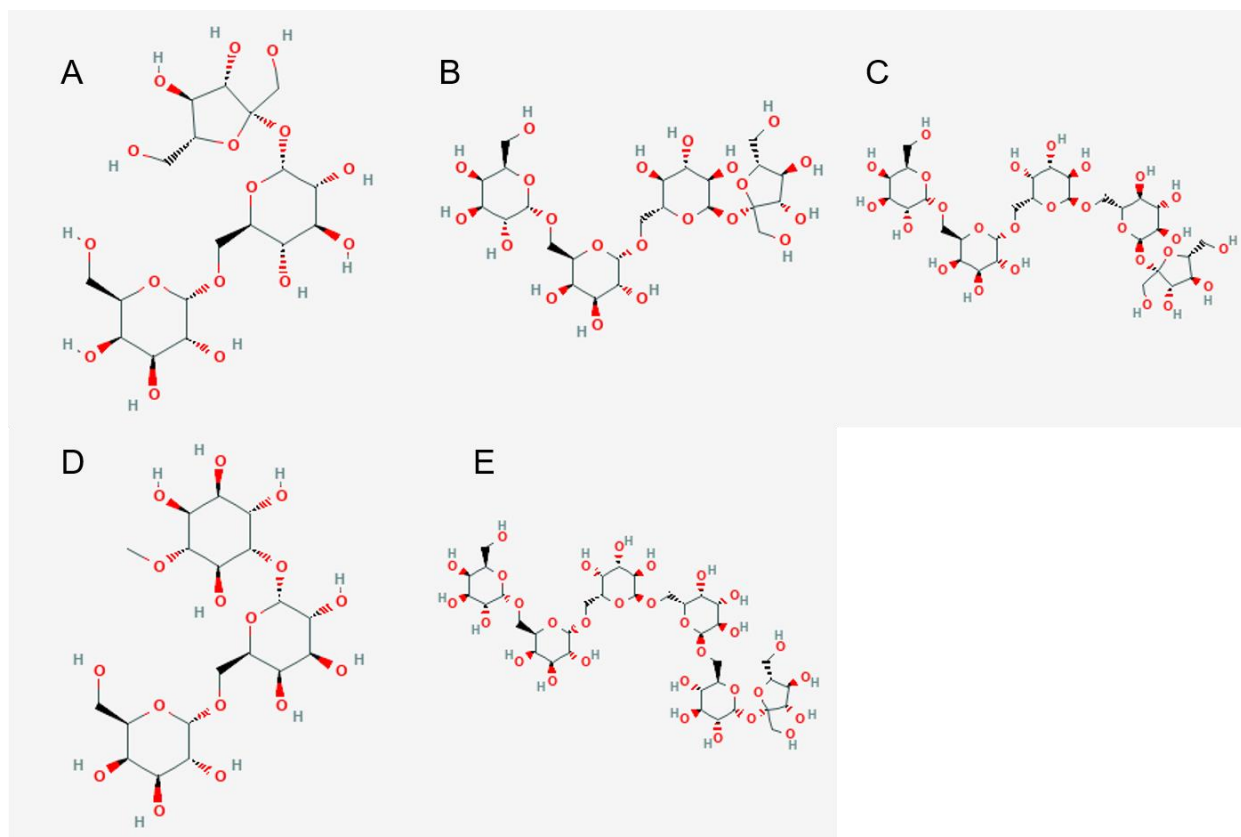


Figure 1.4.1: Examples of raffinose-family oligosaccharides. A – raffinose (MW = 504.4 g/mol); B – stachyose (MW = 666.6 g/mol); C – verbascose (MW = 828.7 g/mol); D – ciceritol (MW = 518.5 g/mol); E – ajugose (MW = 990.9 g/mol). Source: PubChem, 2020.

Ciceritol differs from the rest of the group, since it is a galactosyl cyclitol; each of the cycloalkanes within it, contain at least three hydroxyl groups. However, ciceritol - like all the other members of the raffinose family of oligosaccharides - contains $\alpha(1,6)$ galactosides linked to the C-6 of the glucose moiety of sucrose (Dey, 1980).

These sugars are large molecules, and in general, cannot be absorbed by the stomach (McCleary et al., 2006). They pass undigested through the stomach and into the intestines. The small intestine of monogastric organisms, such as pigs and humans, does not produce the enzyme necessary to convert these complex sugars into their simpler constituent sugars (Suarez et al., 1999).

Thus, these undigested sugars are fermented by colonic bacteria, such as *Escherischia coli*, *Enterococcus faecium*, *Streptococcus macedonius*, *Streptococcus pateuranius*,

Enterococcus avium, etc. These bacteria can either act upon raffinose-family oligosaccharides by hydrolyzing them internally, or they can do so externally.

i) Intracellular hydrolysis: a raffinose transporter is used to bring the raffinose into the cell, and then hydrolysed within the cell to provide energy.

ii) Extracellular hydrolysis: Extracellular fructosidases and levansucrases are used to hydrolyze the raffinose into melibiose and fructose, before the simpler sugars are transported into the cell. (Mao et al., 2018)

Both these processes result in large amounts of carbon dioxide and hydrogen being released into the colon, which can cause bloating, cramping, and flatulence. (Suarez et al., 1999)

1.5 Action of α -galactosidase on raffinose family oligosaccharides

In order to avoid gastric distress, bloating and flatulence, the enzyme α -galactosidase would be required to break down RFOs within the stomach, thereby removing the need for bacterial fermentation. However, as mentioned above, the human digestive system does not produce it.

Acting together with invertase (or β -fructosidase), α -galactosidase breaks down large sugars, such as verbascose, stachyose, raffinose, ajugose and ciceritol into their constituent sugars (which are glucose, galactose, and fructose). It acts upon the α -1,6-links between the monosaccharides (Katrolia et al., 2014), while invertase acts on 1,4-glycoside linkage of sucrose, breaking it down into D-Glucose and D-fructose. (Anilkumar et al., 2017)

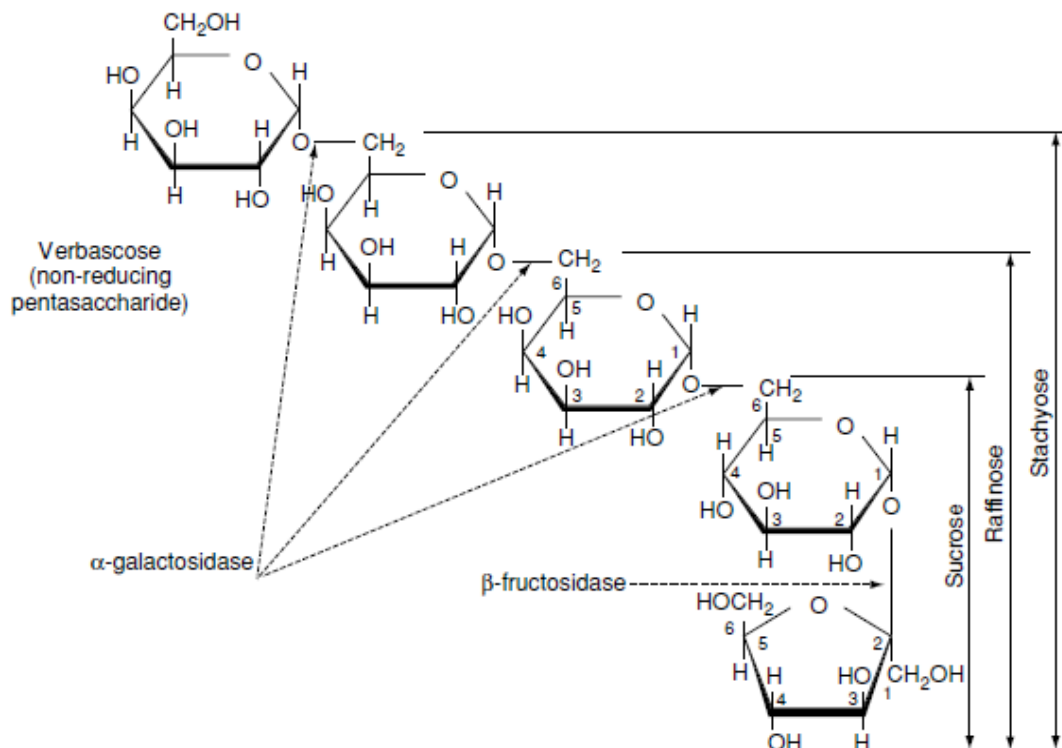


Figure 1.5.1: Relationship between the simpler saccharides present in the raffinose-family oligosaccharides. Figure also indicates the positions at which the enzymes act, to break the bonds connecting the monosaccharides. (Tester and Karkalas, 2003)

Figure 1.5.1 shows the action of α -galactosidase on the α -1,6-links on the saccharides raffinose, stachyose and verbascose. Similarly, the enzyme also functions on these links in ajugose and ciceritol.

Human α -Galactosidase Catalytic Mechanism

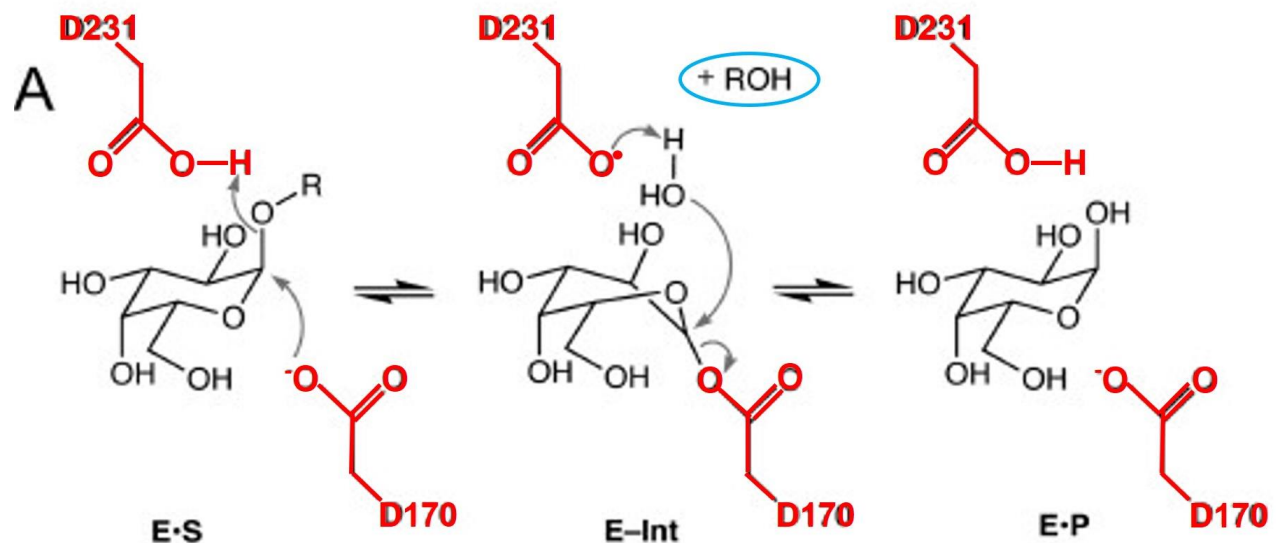


Figure 1.5.2: Activity of α -galactosidase on the substrate, showing the hydrolysis of the α -1,6-linked galactose residues (Guce et al., 2009).

Two parts of the same enzyme attack the substrate simultaneously. These parts, which are aspartic acid residues, are marked in red in figure 1.5.2. This reaction has three stages - the enzyme-substrate stage, the enzyme-intermediate stage, and the enzyme-product stage. These stages are described below:

i) In the Enzyme-Substrate (ES) phase, the two aspartic acid residues from the enzyme, act upon two bonds in the saccharide. A double displacement reaction takes place at the α -1,6 link.

The aspartic acid residue D231 (Asp-231) first acts as an acid, donating a proton. This allows the R-OH to separate from the rest of the sugar. The remaining positive charge is attracted to the other aspartic acid residue, D170 (Asp-170), which behaves as a nucleophile.

ii) In the Enzyme-Intermediate (E-Int) state, it is seen that the R-OH is separated, but the enzyme is still bonded with the intermediate. Residue D231 now behaves as a base, accepting the proton from H-OH, leaving the -OH free to bond with the positive charge now left at the position where -OR used to be.

iii) Finally, in the Enzyme-Product (EP) state, it is seen that the enzyme is now separated from the sugar, and the link between the two sugars was broken and replaced by a proton, thus leaving the monosaccharide in a stable configuration. (Guce et al., 2009)

1.6 Possible solutions to flatulence caused by legume-consumption

As stated previously, certain cuisines do include legumes in their traditional recipes. The issue of flatulence - particularly that caused by the consumption of legumes - is addressed by various means of cooking, in order to reduce or eliminate the RFOs present in them.

For instance, in the work of Larijani *et al.* (2016), the prevention and treatment of flatulence is examined from the perspective of traditional Persian medicine. The research shows that Persian medical scholars have attempted to solve this problem since 1027 CE. Traditional prevention methods include “thoroughly chewing food” and “avoiding drinking beverages during or immediately after the meal”. Treatment options include the consumption of “easily digestible foods, such as roasted chicken and low-fat soups and stewed foods”, as well as consuming a mixture of ground black cumin and honey. This work goes on to show that the recommended herbs (such as caraway, black cumin, ginger, anise, etc.) are used in modern medicine to treat gastrointestinal symptoms. However, in vitro effect of such herbs on raffinose and related sugars is not well documented.

In the work of Song and Chang (2006), several ‘home methods’ were tested against the enzymatic method. These methods include soaking, boiling and autoclaving (pressure cooking), which are methods that can be - and are - employed in the standard household kitchen.

Their work showed that:

- i. Soaking for 16h at room temperature reduced raffinose oligosaccharides by 9.8%
- ii. Boiling for 30, 60 and 90 minutes effectively reduced raffinose oligosaccharides by 44.4%, 46.6% and 52.4% respectively.

By comparison, their work shows that treatment with crude extracellular α -galactosidases from *C. cladosporides*, *A. oryzae*, and *A. niger* (at optimum conditions for each strain) for 3 hours, reduced raffinose and stachyose content by 100% in chickpea flours.

1.7 Enzymatic methods of digesting flatulence factors

The goal of enzymatic digestion of the RFOs, is to prevent the sugars from reaching the colonic bacteria which cause the gas-buildup. Thus, the removal of the sugars from the legumes can occur in two distinct stages: before consumption, or after consumption. If the RFOs are digested after consumption, then this must occur in the stomach, before the biomass can reach the colon.

- i) Pre-digestion of the RFOs: the legumes are milled into a flour and treated with the enzyme. The product that is eaten by the consumer contains very little or no RFOs.
- ii) Consumption of an oral solution of α -galactosidase with flatulence-inducing meals: the consumer eats legumes as per normal, taking an oral solution of the enzyme with their meal. The enzyme reacts with the sugars in the stomach, digesting most or all of the RFOs.

Potential 'market product' forms of the enzyme-treated legumes are either pastes or powders. This is due to the fact that grinding the legumes into a powder before treatment would maximise the reduction of RFOs. Legumes have the property of 'hardseededness' or 'physical dormancy', wherein a water-impermeable seed coat is developed (Smýkal et al, 2014). The seed coat, if undamaged, might protect the contents of the bean from

enzymatic digestion. Thus, for maximum efficiency, milling legumes into a flour for treatment is essential; however, this would call for a change in the style of cooking/eating the legumes, which may influence consumers' decisions.

Over-the-counter solutions of α -galactosidase, such as Beano, are marketed commercially towards individuals that might suffer from flatulence due to the consumption of beans (among other foods). This type of product allows the individual to eat legumes as per their standard habits, and they can simply consume the enzyme with the first bite of their food. In this case, the enzyme reacts with the RFOs in the stomach; the sugars do not pass into the colon for bacterial fermentation, thus avoiding gas buildup, bloating and flatulence.

1.8 Structure of this work

Based on the background information outlined in this section, this work seeks to compare two methods of reducing flatulence by reducing or removing the raffinose family oligosaccharides present in the samples.

The following sections describe the aim of the work, followed by the materials and methods, wherein the processing of the samples is described in detail. The result section then states the changes observed after the treatments; these results are discussed in the detail in the following section, contextualizing the data with regard to legume consumption in daily life, as well as the environmental impact.

The work is summarised in the abstract. The shortcomings of the experiment design are documented in the outlook section. This section ends the study, briefly discussing the conclusions drawn and viewing them through a critical lens.

2. Aim of the work

The meat industry contributes to greenhouse gas emissions and other pollutants. Alternatives to meat - such as veganism, 'fake meat', or legumes - can be used to reduce or completely replace the meat consumption in one's diet. Particularly in the case of legumes, social issues and biological inconveniences can discourage consumers from adding them to their diets. In particular, the consumption of various types of beans is linked with increased flatulence, bloating, cramping and pain.

This problem has been addressed in the past using two main methods:
i) the pre-digestion of the flatulence factors, or
ii) the consumption of an oral solution of α -galactosidase.

In the case of pre-digestion, studies tend to focus on one legume and focus the process on optimizing the digestion for that legume. It is well established that α -galactosidase derived from *A. niger* does reduce the amount of RFOs in a specific sample such as pinto bean or chickpea (Song and Chang, 2006; Mansour and Khalil, 1998). Studies such as these optimize the process to the legume in question. This experiment, however, applies the standardized method of pre-digestion, using α -galactosidase derived from *A. niger*, to a variety of legumes in order to evaluate the effectiveness of this method. If a standard method works across a variety of legumes, then the effort to commercialize 'flatulence-free legumes' will be aided.

In the latter case, studies regarding Beano (Ganiats et al., 1994; Kligerman, 1999) draw data from questionnaires and patient experiences. This work seeks to quantify the extent to which Beano does reduce flatulence factors, extrapolating the data to understand the extent to which it reduces flatulence.

The study was guided by the following questions:

- i. Does enzymatic digestion using α -galactosidase (derived from *Aspergillus niger*) work consistently on many types of legumes to significantly reduce the amount of raffinose family oligosaccharides present in them?
- ii. To what extent does taking an oral supplement of α -galactosidase (such as Beano) reduce the amount of flatulence factors in consumed legumes?
- iii. Does one method outweigh the other in terms of practicality and usability in context of commercialization of flatulence-free legumes?

3. Materials and Methods

Two methods of enzymatic digestion are tested in these experiments: i) pre-digestion of the RFOs in legume flours, and ii) reaction of enzymatic supplement with legumes in the stomach.

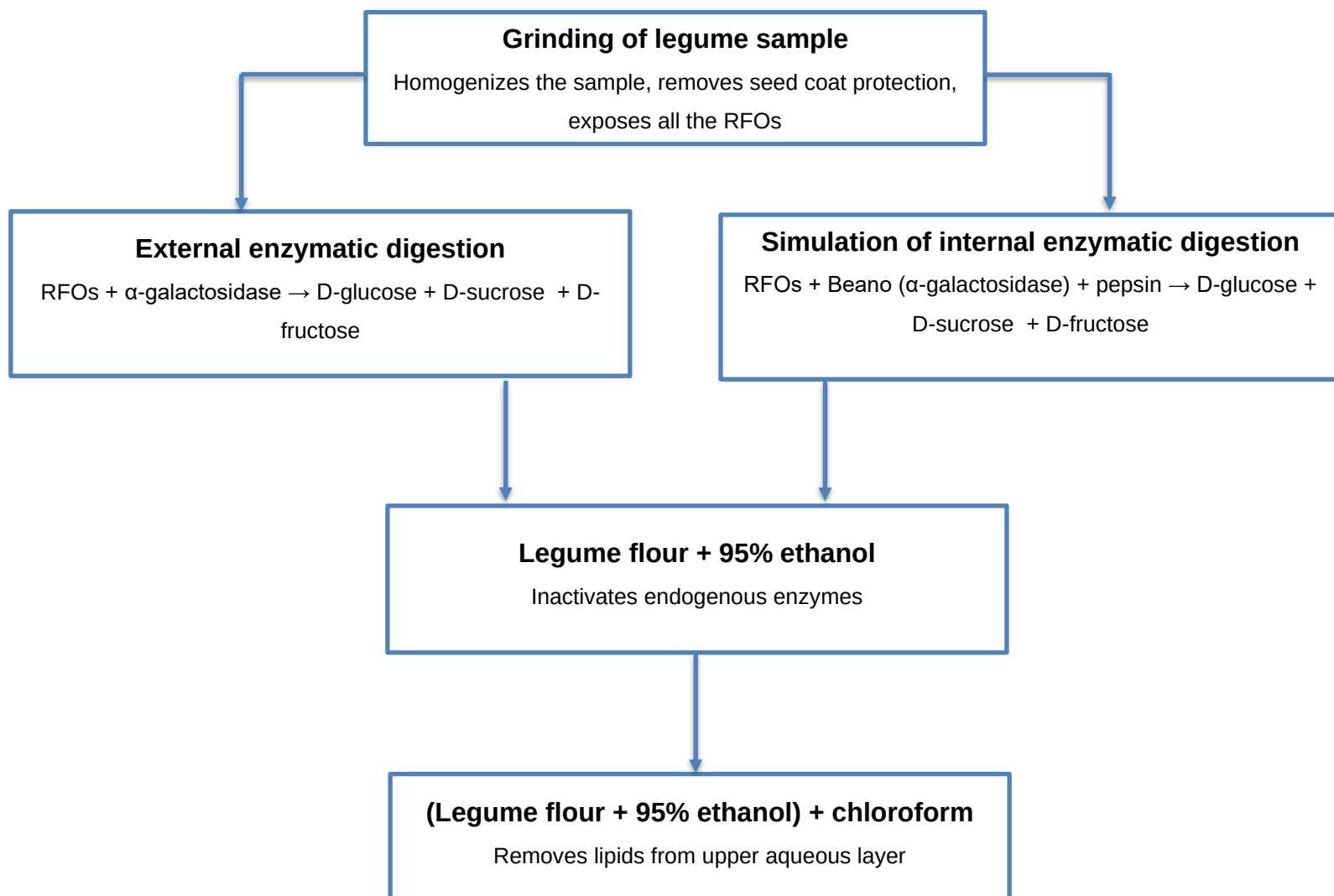
For this study, 6 varieties of legumes were procured from a local grocery store, as noted in table 3.1. The legumes were milled into flour and homogenized by passing through a sieve.

Table 3.1: Legumes used in this work

Legume flour	Identification, binomial nomenclature
Kidney bean	<i>Phaseolus vulgaris</i>
Chickpea	<i>Cicer arietinum</i>
Split pea	<i>Pisum sativum</i>
Yellow lentil	<i>Lens culinaris</i>
Red lentil	<i>Lens culinaris</i>
Green lentil	<i>Lens culinaris</i>

A sample from each legume was treated using the two methods of digestion mentioned. An equal quantity of the legume flour was untreated, as the control factor. The treated and untreated samples were assayed using the Raffinose/Sucrose/D-Glucose Assay Kit from Megazyme International Ireland Ltd.

This assay involves the breakdown of RFOs into sucrose and d-glucose. One mole of each of the RFOs contains one mole of D-glucose (Megazyme International, 2018). The quantity of RFOs was calculated based on the amount of d-glucose present in the samples. Comparing the quantity of RFOs present in each sample shows the amount of reduction for each sample, for both methods. Figure 3.1 demonstrates the process with corresponding chemical interactions.



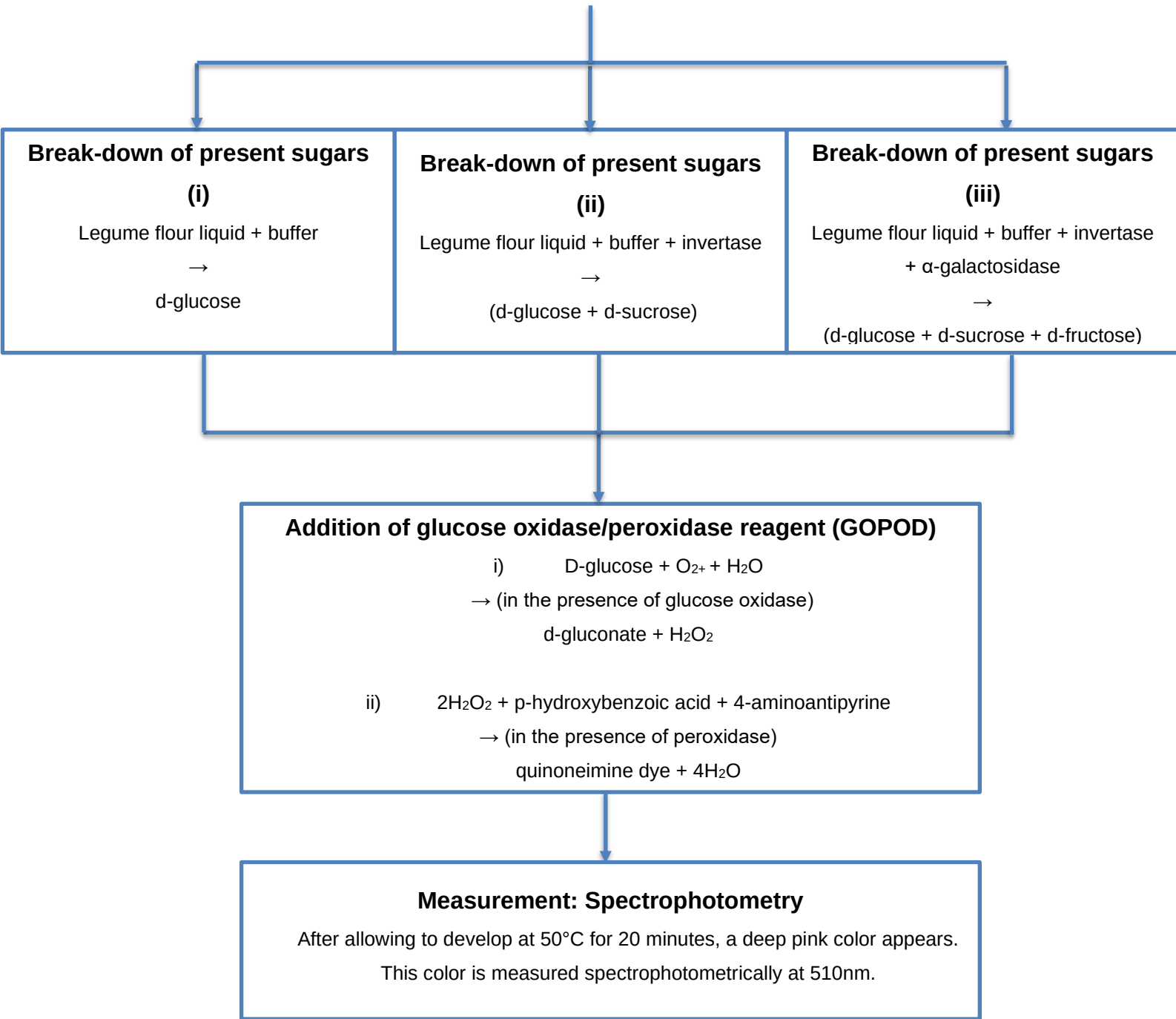


Figure 3.1: Flowchart depicting processing of the sample. Chemical interactions between RFOs and relevant enzymes are shown. Quantification of amount of RFOs is also explained.

This section describes the methods of the following procedures:

3.1. Pre-digestion of flatulence factors

3.2. Digestion of flatulence factors in a simulated stomach environment

3.3. Assay and measurement of flatulence factors

3.1 Pre-digestion of flatulence factors

This protocol is adapted from the other works of research about removing oligosaccharides from legume flours (Mansour and Khalil, 1998; Song and Chang, 2006). For this method, one sample each of all the legume flours was treated with the enzyme, while an equal quantity of each of the legumes was not treated. Treated and untreated samples were both measured for amount of RFOs present. A total of 2 grams of the legume flour is treated, so as to ensure that there is enough digested flour for a triplicate of the assay.

3.1.1. Materials used

Table 3.1.1: Materials used for pre-digestion

Materials	Specifics
Legume flour	6 legumes x (1 digested test + 1 undigested test) x 2 g of each sample Total = 24 g of legume flour
α -galactosidase, 60 U/mL	10 mL per (digested) gram of sample = 20mL per sample = 6 samples x 20 mL Total = 120 mL
Water bath	40° C
Whatman no. 1 filter papers	-
Dessicator	40° C

3.1.2 Protocol

10 mL of α -galactosidase and 1 gram of legume flour were added into a reaction tube and mixed well. The reaction mixture was incubated in a water bath at 40° for 1 hour, and the rack containing the reaction tubes was manually agitated every five minutes.

The mixture was then filtered through Whatman no. 1 filter papers and the insoluble solids were dried for 4 hours at 40°. The solids were homogenized again by grinding them to produce the enzyme-treated flour.

3.2 Digestion of flatulence factors in a simulated stomach environment

To test the enzyme-supplement method, the supplement “Beano” was used. It was procured online (at https://www.amazon.co.uk/Beano-Gas-Relief-Digestion-Tablets/dp/B01F9DT5GM/ref=sr_1_5?). More information about the product can be found in the appendix (section 8.1). This supplement was reacted with each of the legume flours in a simulated stomach environment. For control, an equal quantity of the flour was incubated in the simulated stomach environment without being treated with Beano. Both samples, treated and untreated, were tested for amount of RFOs present.

3.2.1 Materials used

Table 3.2.1: Materials required for digestion in stomach environment.

Material	Specifics, per sample
Zippered plastic bag	250 mL volume, 1x
Graduated cylinder	50 mL, 1x
Water bath	37°C

Table 3.2.2: Reagents required for digestion in stomach environment.

Reagent	Quantity per sample
Legume flour	37 g
α -galactosidase, 800GAL/U	2 tablets of Beano
Pepsin; 0.5% solution	50 mL
Hydrochloric acid, 0.001 M, pH = 3	50 mL

3.2.2 Protocol for constructing the simulated stomach environment

This protocol was adapted from the one described in The American Biology Teacher, under the How To Construct An Artificial Stomach (Culp, 2010) chapter, as well as protocols described in other relevant research works which used simulated stomach environments (Jung & Lee, 1998; Purchas et al., 2006).

The water bath was set up at 37°C. 50 mL of the HCl solution was placed in the zippered bag, followed by 50 mL of the pepsin solution. This bag, when sealed and placed in the beaker in the water bath, functioned as the artificial stomach.

Gastric lipase, the other enzyme present in the stomach, was omitted due to limited resources, as well as the low lipid quantities in the legumes sampled.

3.2.3 Digestion of flatulence factors in the simulated stomach environment

This protocol is adapted mainly from research about the nitrosation of food in a simulated gastrointestinal environment (Newton, 1975).

37g of the legume flour, assumed to be one serving of legumes, was added to the simulated stomach environment (Kantor, 1998). The contents were agitated manually to

mix the flour with the 'stomach acid'. 2 tablets of Beano (containing 800 U of the enzyme), were crushed up and added immediately after this. Usage recommendations state that 2 tablets are to be taken with the meal (see Appendix 8.2). The US patent for Beano states that dosages under 675 U tend to be minimally effective (Kligerman, 1999). Therefore, 800 U of the enzyme is accepted as an appropriate amount for one serving, 37 g.

The mixture was stirred to ensure homogenization, and then incubated for 300 minutes (gastric emptying time as per Camilleri et al., 1989) in a water bath at 37°C. The pH was tested every 30 minutes and adjusted to 3 as required, using HCl. The reaction product was centrifuged at 12,500 x g for 20 minutes; the supernatant was decanted and discarded. The solids were washed with water three times. They were then filtered and dried at 40°C for 4 hours.

3.3 Assay and Measurement of flatulence factors

This protocol follows the one provided with the pre-made kit procured from Megazyme Ltd. (Megazyme International, 2018).

It follows the principle that RFOs are hydrolysed to d-galactose, d-glucose, and d-fructose using α -galactosidase and β -fructosidase (invertase). The quantity of d-glucose is determined using the glucose oxidase/oxidase (GOPOD) reagent.

The quantification of glucose using the GOPOD reagent can be done colorimetrically, and the samples are examined spectrophotometrically at 510 nm.

The following samples are measured:

- i) a blank reagent
- ii) a standard solution of d-glucose
- iii) an untreated flour sample
- iv) a treated flour sample.

3.3.1 Materials used

Table 3.3.1: Equipment and reagents required for assay and measurement

Equipment	Specifics
Water bath	84 - 88°C
Volumetric flask	50 mL
Spectrophotometer	Set at 510 nm
Vortex mixer	-
Centrifuge	1000 x g

Reagents	Quantity per test
Ethanol, 95% solution	5 mL
Chloroform	2 mL
Legume residue	0.5 g
Buffer I (50mM sodium acetate)	50 mL + 0.4 mL = 50.4 mL
Invertase + α -galactosidase suspension	0.2 mL
Invertase solution	0.2 mL
Reagent blank (Buffer I)	0.4 mL
Glucose control	0.1 mL D-Glucose standard solution + 0.3 mL Buffer I
Glucose determination reagent (GOPOD reagent)	(3.0 mL x 5 solutions =) 15.0 mL

3.3.2 Protocol

0.5 gram of the sample flour and 5 mL of 95% ethanol were added into a 15ml reaction tube. The mixture was incubated in an 84°C water bath for 5 minutes. This treatment inactivated endogenous enzymes. The mixture was transferred to a 50ml reaction tube and the volume was adjusted to 50ml with sodium acetate buffer. This mixture was capped and allowed to sit for 15 minutes, and then shaken vigorously.

5 ml of the solution was transferred to a centrifugation test tube, and 2 ml of chloroform was added to it. This tube was vortexed for 15 seconds, and then centrifuged at 1000g for 10 minutes. Treating the solution with chloroform removed most lipids from the upper aqueous phase. The upper aqueous layer from the tube was extracted. 0.2mL aliquots of this upper aqueous layer was added to 0.2mL each of the following:

- i) Buffer I,
- ii) invertase solution and
- iii) invertase + α -galactosidase solution, each.

These were incubated at 50°C for 20 minutes.

3 mL of GOPOD Reagent was added to all three solutions, as well as to the standard and the D-Glucose control. All 5 solutions were incubated at 50°C for 20 minutes. The absorbance for each was read at 510nm.

3.3.3 Calculation of flatulence factors

The amount of raffinose family oligosaccharides present was calculated using the procedure given in the assay protocol (Megazyme International, 2018). The amount of simple sugars in the sample was calculated as a control factor. Further, complex sugars were enzymatically broken down into simple sugars. The difference between the control figure and the samples shows the quantity of RFOs present. In all the calculations, the following factors are used:

Table 3.3.3.1: Explanation of factors used in the calculations.

Factor	Explanation
ΔA	Absorbance for a sample containing the flour + acetate buffer + 3.0 mL of GOPOD Reagent
ΔB	Absorbance for a sample containing the flour + invertase + 3.0 mL of GOPOD Reagent
ΔC	Absorbance for a sample containing the flour + α-galactosidase and invertase + 3.0 mL of GOPOD Reagent
F	a factor to convert from absorbance to μmoles of glucose $= \frac{0.556 \text{ } \mu\text{moles of glucose}}{\text{GOPOD absorbance for } 0.556 \text{ } \mu\text{moles of glucose}}$
1/1000	Used to convert from micromoles to millimoles
250	Used to convert to 50 mL of extract from 0.2 mL
200	Used to convert from 100 g of sample from 0.5 g
10	Used to convert from grams/100grams to milligrams/gram

i) Quantity of D-glucose:

$$Q_{D-Glucose} \frac{\text{millimoles}}{100 \text{ g flour}} = \Delta A \times F \times 250 \times 200 \times \frac{1}{1000} = \Delta A \times F \times 50$$

This can be further quantified in terms of milligrams of D-Glucose per gram of bean flour by:

$$= Q_{D-Glucose} \frac{\text{millimoles}}{100 \text{ g flour}} \times 180.16 \frac{\text{g}}{\text{mol}} \times \frac{1}{1000 \text{ mg D-Glucose}} \times 10$$

Where $180.16 \frac{\text{g}}{\text{mol}}$ is the molecular weight of D-glucose (NCBI).

ii) Quantity of sucrose

$$Q_{\text{Sucrose}} \frac{\text{millimoles}}{100 \text{ g flour}} = (\Delta B - \Delta A) \times F \times 250 \times 200 \times \frac{1}{1000} = (\Delta B - \Delta A) \times F \times 50$$

This can be further quantified in terms of milligrams of D-Glucose per gram of bean flour by:

$$= Q_{\text{Sucrose}} \frac{\text{millimoles}}{100 \text{ g flour}} \times 342.3 \frac{\text{g}}{\text{mol}} \times \frac{1}{1000 \text{ mg sucrose}} \times 10$$

Where $342.3 \frac{\text{g}}{\text{mol}}$ is the molecular weight of sucrose (NCBI).

iii) Quantity of raffinose-family oligosaccharides (RFO), millimoles/100 grams:

$$Q_{\text{RFO}} \frac{\text{millimoles}}{100 \text{ g flour}} = (\Delta C - \Delta B) \times F \times 250 \times 200 \times \frac{1}{1000} = (\Delta C - \Delta B) \times F \times 50$$

The total quantity of Raffinose-family oligosaccharides in the flour cannot be quantified in the same manner as done for D-Glucose and Sucrose, since these oligosaccharides contain a mixture of raffinose, stachyose and verbascose. It is possible to use the molecular weight of the most prevalent component, if known. The following equation gives the amount of RFOs present in the sample in milligrams per gram.

$$= Q_{\text{RFO}} \frac{\text{millimoles}}{100 \text{ g flour}} \times \text{molecular weight of most prevalent oligosaccharide} \frac{\text{g}}{\text{mol}} \times \frac{1}{1000 \text{ mg of the relevant saccharide}} \times 10$$

3.4 Statistical analysis

All data is represented as mean $\pm$ standard deviation unless otherwise mentioned.

The collected data was organized in Microsoft Excel (MS Office 365)/Google Sheets. Since there was no continuous independent variable, no linear regression tests were conducted. The data sets were tested for normality using the Shapiro-Wilk test and QQ plots, using GraphPad Prism (version 8.4.0 (671)).

QQ plots were assessed visually, and showed that the data sets were normal. Thus, repeated measures one-way ANOVA tests were conducted on each of the data sets (6 total), in order to establish whether or not any α -galactosidase treatment is significantly effective. The test was validated by checking for the normal distribution of residuals.

Post hoc, Tukey's multiple comparison testing was done to determine which treatments were significantly effective. A p value <0.05 was considered significant.

All data was plotted using GraphPad Prism (version 8.4.0 (671)).

4. Results

RFO content present in each legume tested was determined as described in section 3.3.3c. Among all legumes tested, yellow lentils and kidney beans have the lowest RFOs contents (Table 4.1). Table 4.1 shows the amount of RFOs found in untreated samples of the legumes. As seen in section 3.3.3.c, the quantification of RFOs in each sample required the molecular weight of the oligosaccharide with the greatest concentration, in the legume. Thus table 4.1 also shows which RFO was used for the calculation.

Table 4.1: Comparison between amount of RFOs observed in samples and literature values

	Total amount of RFO $\pm$ standard deviation (mgg^{-1})	Most prevalent RFO	Molecular weight of most prevalent RFO (gmol^{-1})	Amount of total RFOs in literature (mgg^{-1})	Source
Kidney Bean	26.15 ± 3.70	Stachyose	666.58	37.5	McPhee et al., 2002
Chickpea	91.41 ± 4.34	Ciceritol	518.46	144.9	Han and Baik, 2006
Split Pea	69.40 ± 7.25	Stachyose	666.58	80.4	Han and Baik, 2006
Red Lentil	90.57 ± 4.69	Ciceritol	518.46	122.9	Han and Baik, 2006
Yellow Lentil	23.77 ± 1.78	Ciceritol	518.46	23.6	Kleintop et al., 2013
Green Lentil	88.79 ± 12.94	Ciceritol	518.46	95.5	Han and Baik, 2006

As described in sections 3.1 and 3.2, two different methods of digestion were applied to all the samples (in triplicate). By comparison to the control samples, reduction in the amount of RFOs in the sample were seen with each treatment, including digestion without Beano. However, significant reduction is only seen in the pre-digestion method, and with the Beano treatment. Table 4.2 shows the amount of RFOs present in each sample after the three treatments (pre-digestion, digestion without Beano, and digestion with Beano). These figures are compared with the amount of RFOs present in the control sample.

Table 4.2: Effect of various treatments on each of the samples.

	Control (mgg ⁻¹ ± SD)	Pre-digested (mgg ⁻¹ ± SD)	Digested without Beano (mgg ⁻¹ ± SD)	Digested with Beano (mgg ⁻¹ ± SD)
Kidney Beans	26.16 ± 3.70	2.26 ± 0.42*	21.54 ± 1.77	6.82 ± 0.61*
Chickpeas	91.41 ± 4.34	4.64 ± 0.25*	74.15 ± 3.73	4.80 ± 1.40*
Split Peas	69.40 ± 7.25	3.01 ± 0.88*	58.92 ± 1.50	4.78 ± 0.81*
Red Lentils	90.57 ± 4.69	6.48 ± 2.20*	70.35 ± 1.80	8.15 ± 1.15*
Yellow Lentils	23.77 ± 1.78	2.88 ± 1.82*	17.30 ± 1.75	4.76 ± 2.54*
Green Lentils	88.80 ± 12.94	4.14 ± 1.13*	68.31 ± 2.94	7.52 ± 4.71*

Each value is the mean ± standard deviation for 3 replicates.

* indicates that the value is significant at P < 0.05 according to Tukey's HSD test.

4.1 Effectiveness of pre-digestion across various types of legumes

In order to evaluate the effectiveness of the pre-digestion method, the quantity of RFOs before and after treatment is compared. Quantifying the reduction in the form of percentage further supports the visualization of this data. Similar results are seen for all the legumes, ranging from 91% to 97% (as seen in table 4.1.1).

Table 4.1.1: Effect of pre-digestion treatment on various legumes.

	Average amount of RFO in control (mgg ⁻¹)	Average amount of RFO in pre-digested sample (mgg ⁻¹)	Reduction seen (%)
Kidney Bean	26.16 ± 3.70	2.26 ± 0.43	91.38*
Chickpea	91.41 ± 4.34	4.64 ± 0.25	95.19*
Split Pea	69.40 ± 7.25	3.01 ± 0.89	94.33*
Red Lentil	90.57 ± 4.69	6.48 ± 2.20	92.95*
Yellow Lentil	23.77 ± 1.78	2.88 ± 1.82	95.56*
Green Lentil	88.80 ± 12.94	4.14 ± 1.13	97.46*

Each value is the mean ± standard deviation for 3 replicates.

* seen next to a figure in the reduction column indicates that the value is statistically significant according to Tukey's HSD test.

The results seen in table 4.1.1 demonstrated that the pre-digestion method is significantly effective in all the types of legumes tested. Figure 4.1.1 demonstrates the difference in RFO quantity before and after pre-digestion treatment in all the samples.

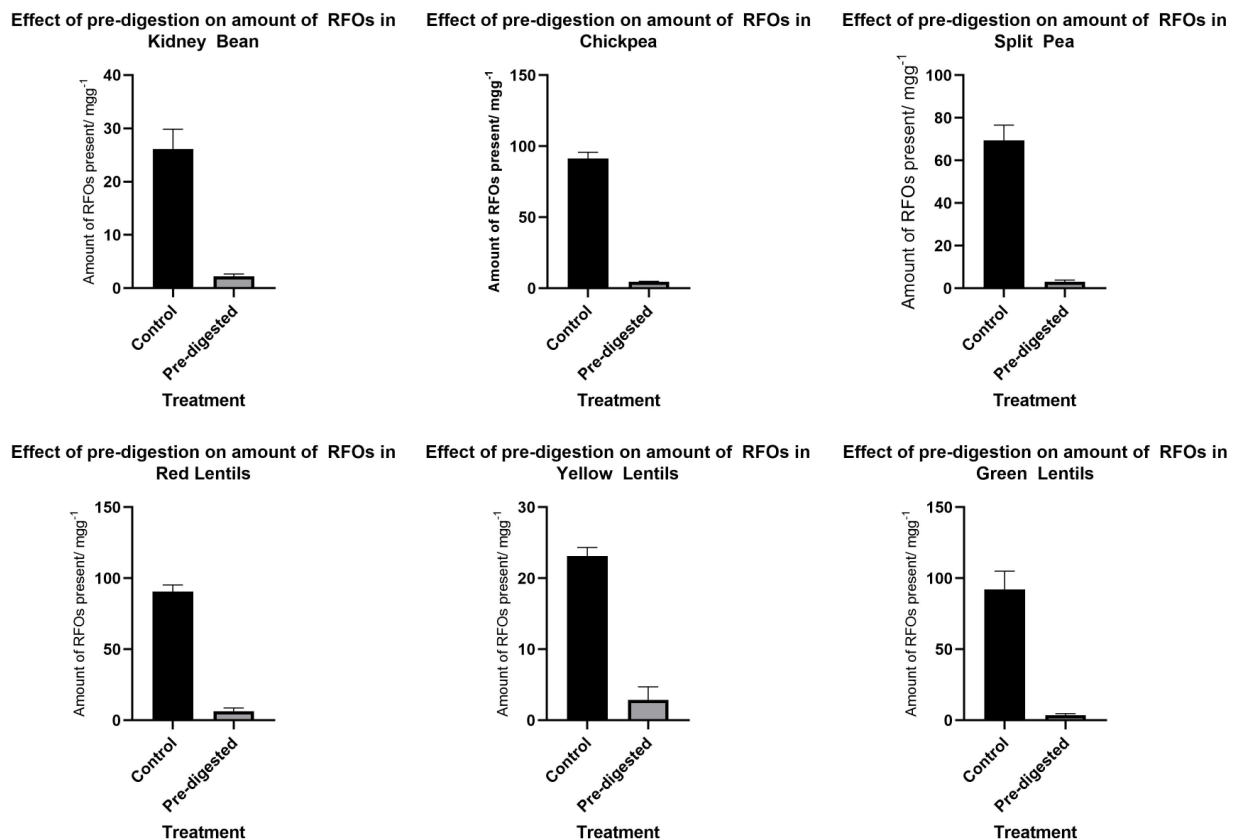


Figure 4.1.1: Effect of pre-digestion on various legumes.

4.2 Extent of RFOs reduction in a stomach environment by using Beano

Effectiveness of the Beano method is established by comparing the results post-digestion, with the control sample. The average amount of RFOs found in the samples digested without Beano are also included, in order to show that a small amount of RFOs does get digested even without Beano. However, that quantity is not at all significant, as seen in table 4.2.1. A greater variance in extent of reduction is seen with this method; results range from 68% (in kidney beans) to 91% (in chickpeas).

Table 4.2.1: Comparison between control sample, digested sample, and sample digested with Beano

	Average amount of RFOs in control sample / mgg^{-1}	Average amount of RFOs in sample digested without Beano / mgg^{-1}	Average amount of RFOs in sample digested with Beano / mgg^{-1}	Extent of reduction of RFOs / %
Kidney Bean	26.16 ± 3.70	21.54 ± 1.77	6.82 ± 0.61	68.34*
Chickpea	91.41 ± 4.34	74.15 ± 3.73	4.80 ± 1.40	91.76*
Split Pea	69.40 ± 7.25	58.92 ± 1.50	4.78 ± 0.81	90.59*
Red Lentil	90.57 ± 4.69	70.35 ± 1.80	8.15 ± 1.15	86.89*
Yellow Lentil	23.77 ± 1.78	$17.30 \pm .75$	2.88 ± 1.82	87.10*
Green Lentil	88.80 ± 12.95	68.31 ± 2.94	7.52 ± 4.71	77.78*

Each value is the mean $\pm$ standard deviation for 3 replicates. * seen next to a figure in the 'extent of reduction' column indicates statistical significance according to Tukey's HSD test.

Table 4.2.1 includes the amount of RFOs in the control sample in order to demonstrate that the digestion process does reduce the amount of RFOs significantly. The comparison that is focused on in this table, is that between digestion with and without Beano.

The process showed good effectiveness across all samples - the RFO content was significantly reduced in all the legumes.

Figure 4.2.1 illustrates the difference between control samples, digested samples, and samples digested with Beano. The figures follow the trend of a small reduction in RFOs upon digestion without Beano, and a much larger reduction upon digestion with Beano.

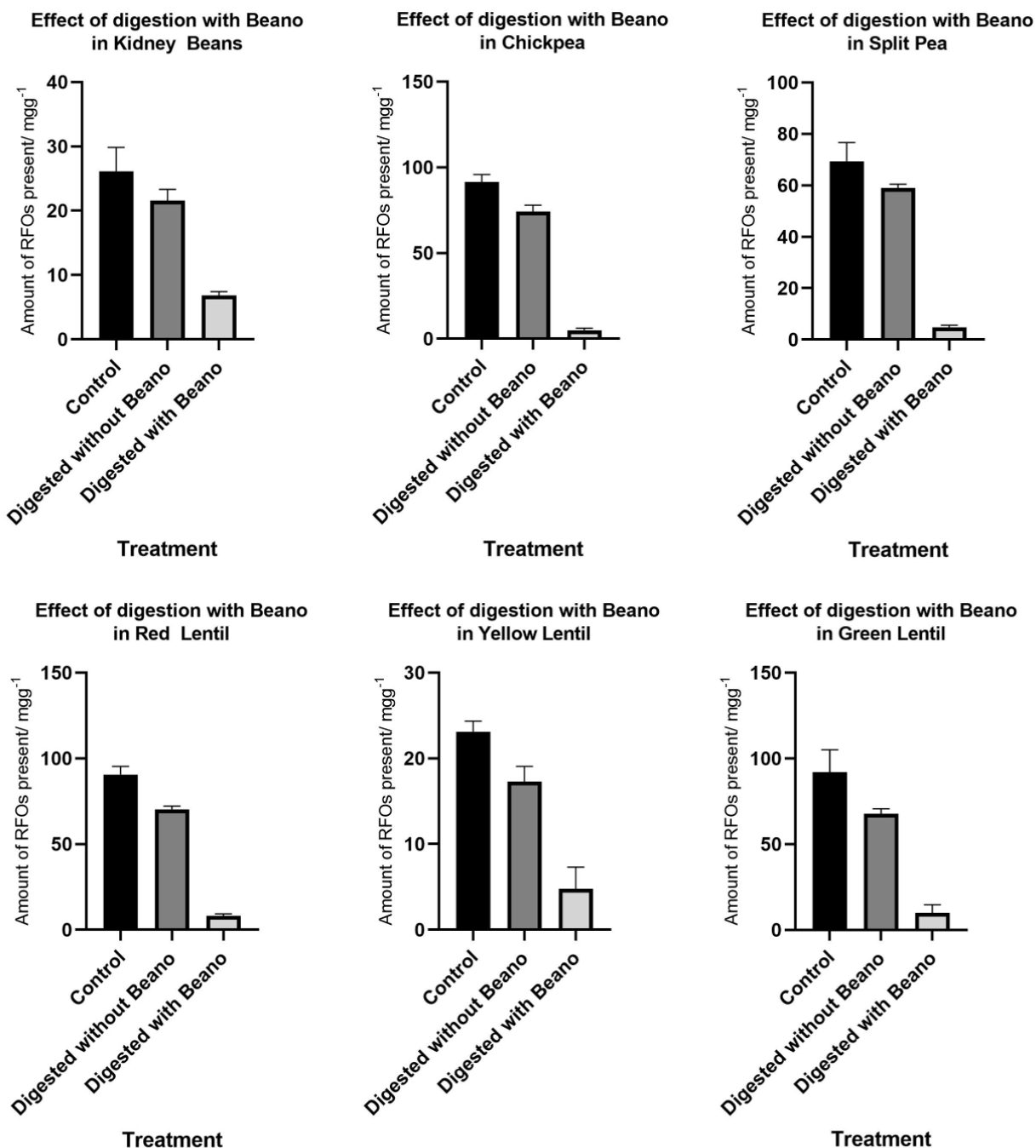


Figure 4.2.1: Comparison between control samples, samples digested without Beano, and samples digested with Beano, in all the legumes tested.

4.3 Comparison between two methods

Having established that both methods do significantly reduce the amount of RFOs across several types of legumes, it is now important to evaluate if one method does so significantly better than the other. In order to do so, the average extent of reduction is compared between both methods, as seen in table 4.3.1. Two samples (kidney bean, yellow lentil) show a significant difference between the two methods.

Table 4.3.1: Difference in extent of reduction of RFOs in samples using both methods

Legume	Treatment	Average extent of change $\pm$ standard deviation (%)
Kidney Bean*	Pre-digestion	91.15* $\pm$ 2.73
	Digestion with Beano	68.21 $\pm$ 3.68
Chickpea	Pre-digestion	94.92 $\pm$ 0.34
	Digestion with Beano	93.55 $\pm$ 1.64
Split Pea	Pre-digestion	95.58 $\pm$ 1.53
	Digestion with Beano	91.89 $\pm$ 1.33
Red Lentil	Pre-digestion	92.76 $\pm$ 2.76
	Digestion with Beano	88.44 $\pm$ 1.38
Yellow Lentil*	Pre-digestion	87.90* $\pm$ 7.90
	Digestion with Beano	73.27 $\pm$ 12.39
Green Lentil	Pre-digestion	96.06 $\pm$ 1.22
	Digestion with Beano	85.16 $\pm$ 7.32

Each value is the mean $\pm$ standard deviation for 3 replicates. * seen next to a value in the 'legume' column indicates that there is a significant difference between the two processes.

The difference between the treatments is not seen to be significant in 4 out of the 6 samples. Figure 4.3.1 demonstrates the difference in extent of reduction caused by the two treatments.

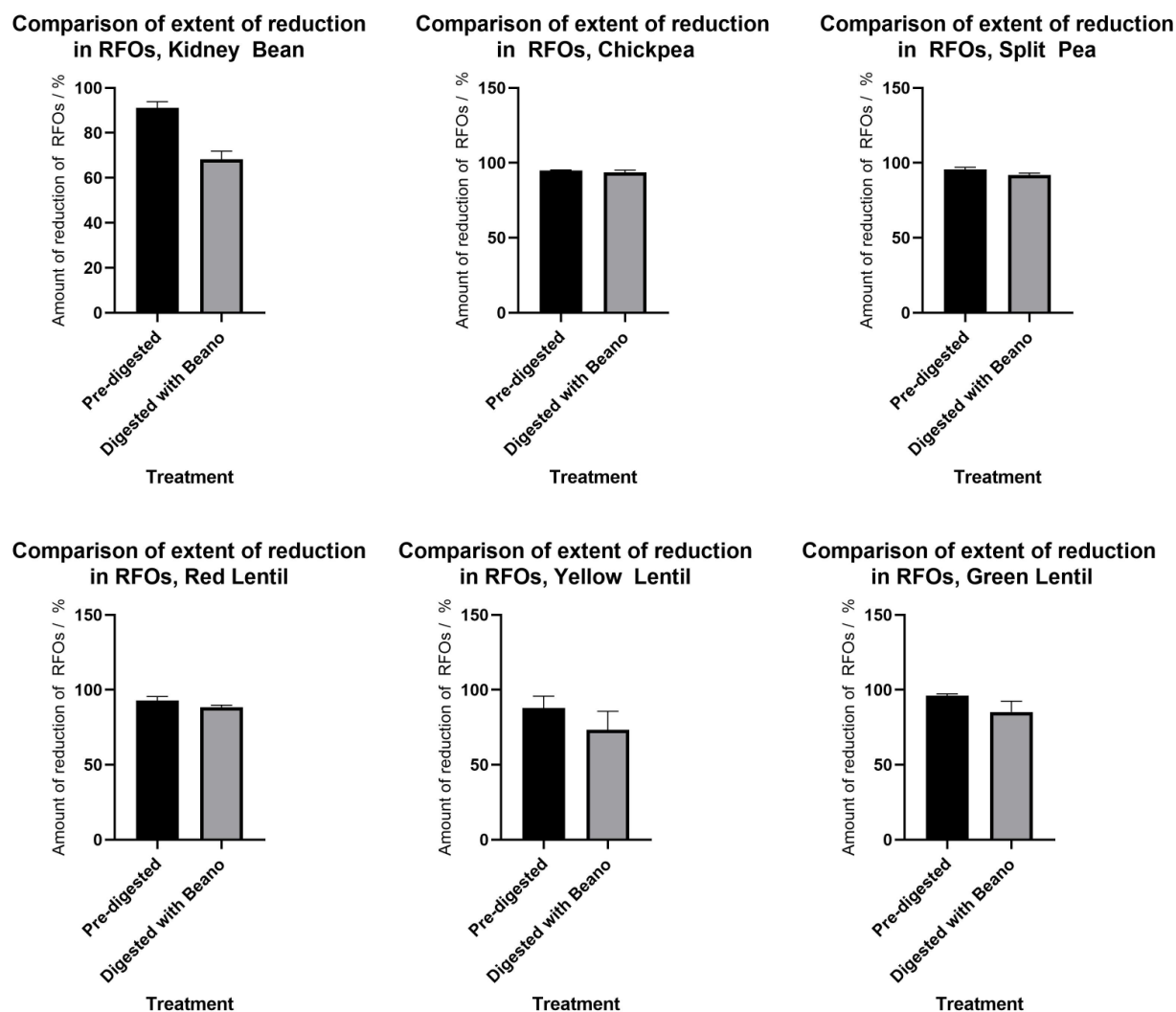


Figure 4.3.1: Comparison between effectiveness of two methods of treatment.

Figure 4.3.1 visually demonstrates the similarity in the effectiveness of the two methods. The two samples which do show a significant difference are kidney bean and yellow lentil.

5. Discussion

This research was guided by three questions; thus this section will address each question in a separate subsection.

5.1. Effectiveness of pre-digestion method on different legumes

Guiding question: does enzymatic digestion using α galactosidase (derived from *Aspergillus niger*) work consistently on various types of legumes to significantly reduce the amount of raffinose family oligosaccharides present in them?

α -galactosidase may be extracted from bacterial, fungal or plant sources. The supplier of the α galactosidase used for this experiment - Megazyme International - offers the enzyme extracted from fungi (*Penicillium simplicissimum*, *Aspergillus niger*), and plants (guar) but not from bacteria. Enzyme extracted from *A. niger* was selected on the basis of the study conducted by Mansour and Khalil (1998), as well as that of Song and Chang (2006), both of which showed that α -galactosidase derived from *A. niger* is able to reduce the quantity of RFOs in legumes by 100%.

Every study used as literature for this experiment, tested the effectiveness of α -galactosidase from different sources on a single substrate (Song and Chang, 2006; Mansour and Khalil, 1998) or the effectiveness of other methods of treatment such as immobilizing the enzyme and testing it on a single substrate (Kotiguda et al., 2007).

The works mentioned above were specialized to focus on the legume of choice whereas this experiment applied a standardized pre-digestion method to a variety of legumes, with varying amounts of RFOs in the control samples. This experiment was conducted to test whether a standardized method would work across several different types of legumes, with varying amounts of RFOs.

Results seen in section 4.1 demonstrate a significant reduction in the amount of RFOs in all the legumes tested. Figure 5.1 compares the extent of reduction in amount of RFOs

between all the samples tested, and the results shown by Song and Chang (2006), and Mansour and Khalil (1998).

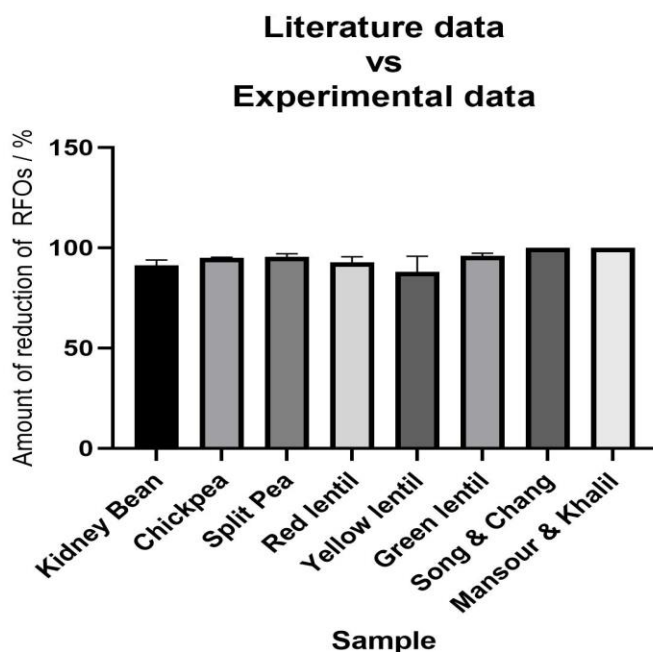


Figure 5.1: comparison between extent of reduction of RFOs in literature and experimental data.

Upon conducting a repeated-measure one way ANOVA for this data set, it was found that none of these figures are significantly different. This supports the hypothesis that treatment with α -galactosidase derived from *A. niger* significantly reduces the amount of RFOs in various types of legumes.

5.2. Extent of reduction of RFOs by using Beano

Guiding question: To what extent does taking an oral supplement of α -galactosidase (such as Beano) reduce the amount of raffinose family oligosaccharides in consumed legumes?

It has been established that in order to prevent the gas buildup that causes bloating, abdominal pain, and flatulence, RFOs must be removed before they can be metabolized by intestinal bacteria. If this step is to be performed after the consumption of a normal meal, then α -galactosidase should be consumed orally, so as to allow the RFOs to be

broken down into smaller sugars (such as sucrose, d-glucose) in the stomach. In this experiment, this situation was simulated with and without the use of the supplement Beano.

Section 4.2 showed that the amount of RFOs in the samples was significantly smaller after digestion with Beano. On average, digestion with Beano reduced the amount of RFOs in the samples by 83.74%.

Other studies about the effect of oral α -galactosidase on intestinal gas used a human testing model (Ganiats et al., 1994; Kligerman, 1999; Di Stefano et al., 2007). In each of these tests, the subjects were given a meal containing RFO-rich foods, absent of lactose-containing components or of soda/alcohol. The subjects were then given either a high dose, low dose or placebo of α -galactosidase. Table 5.2.1 illustrates the differences between the studies. The “Does Beano Prevent Gas” (Ganiats et al., 1994) study is omitted since it does not describe the quantities of food or enzyme precisely.

Table 5.2.1: differences in studies related to Beano/oral supplements of α -galactosidase

	Kligerman, 1999	Di Stefano et al., 2007	This study
Amount of food (grams)	230	420	37
Amount of RFO in food (grams)	5	7.56	0.87 - 5.36
High dose of α -galactosidase (GalU)	2250	1200	800
Low dose of α -galactosidase (GalU)	675	300	800
Amount of α -galactosidase per gram of RFO (of significantly effective treatment) (GalUg ⁻¹)	450	158.73	149.25 - 919.54

Both tests compared above relied on data taken over 8 hours at regular intervals, while the samples in this experiment were tested for the amount of RFOs once, at the end of a 5 hour period. The studies also have similar measurements of symptoms - breath H₂ and self-described symptoms.

In the Di Stefano study (2007), it was seen that 1200 GalU of α -galactosidase significantly reduced breath H₂ excretion, while reduction from a 300 GalU dose was not significant. Subjects also reported on number of flata; the 300 GalU dose did not significantly reduce flata, while the 1200 GalU dose did.

In comparison, for the experiments for the US patent for Beano (Kligerman, 1999), doses of 675 GalU and 2250 GalU were tested. In this case, both doses appear to significantly reduce breath H₂ excretion. Subjective symptoms were reported by test subjects. The 3 mL dose (675 GalU) shows to reduce (but not completely remove) symptoms. The 10 mL (2275 GalU) dose completely removed all symptoms in all subjects.

In both of these studies, the higher dose (1200 U, 2250 U) showed a significant reduction in amount of breath H₂ and symptoms of intestinal gas. In table 5.2.1., the ratio of units of enzyme to grams of RFO in samples was compared. The ratios that produced significant results (158.73 GalU/gram, 450 GalU/gram) correspond to the figures from this experiment, which ranged from 149.25 GalU/gram of RFO for chickpea to 919.54 GalU/gram for yellow lentils.

The discrepancy in the varying ratios of enzyme-to-amount of RFOs in the samples is explained by the difference in amount of RFOs in the untreated samples. All samples were separated into 37 gram portions, and treated with the same amount of Beano (2 tablets, 800 GalU).

Since the ratios of effective treatments from literature do correspond with the figures in this experiment, it is reasonable to assume that an average of 83.74% reduction in RFOs

does significantly reduce symptoms of intestinal gas, such as bloating, abdominal pain and flatulence.

However, it is also important to note the major differences between the experiment design used here, and the human testing model. This model utilized the enzyme pepsin, a protease, in the simulated stomach environment. In the human stomach, lipases would also be excreted. Lipase was omitted due to resources being limited.

It is also important to note that the digestion in this experiment took place with a static amount of 'gastric fluid', and did not attempt to replicate the flows of the digestive system for the sake of simplicity.

5.3. Comparison of two methods

Guiding question: Does one method outweigh the other in terms of practicality and usability in context of commercialization of flatulence-free legumes?

In order to answer this question as fully as possible, several avenues must be explored.

5.3.1 Observed results

It was seen in section 4.3 that out of the 6 samples tested, a significant difference was seen in only 2 of the samples (kidney bean, yellow lentil). It is worth noting that these two legumes did have the lowest amount of starting RFO out of all the samples.

It is reasonable to say that in general, digestion with Beano and pre-digesting with α -galactosidase give relatively similar results.

Viewed through the lens of practicality (from a consumer's perspective), there would be little difference between buying flatulence-free legume flour and taking Beano with meals, except the additional step of purchasing Beano. If the consumer is prone to eating kidney beans or yellow lentils over the other legumes in this experiment, then they would reduce

flatulence more by consuming a pre-digested bean flour rather than by taking Beano with their meal.

5.3.2. Social and logistical benefits and shortcomings of both methods

Oral enzymatic supplements:

- i. Beano and other, more general digestive supplements, are already sold globally as digestive agents. Thus, the consumer may be more likely to choose a well-established product such as this one, over a flatulence-free legume flour, with which they are not familiar.
- ii. The consumer need not change their cooking habits if they are taking Beano. If, however, they prefer pre-digested legumes, then they must change their style of cooking to suit this alternate form.
- iii. Since this study is contextualized in the Netherlands, it is also important to evaluate the usability of Beano here. It is not commercially available, and for this study, it had to be ordered from England. The process was not time consuming, but a regular consumer may not want to expend extra time and money on international delivery of Beano. It is worth noting that generic, multi-enzyme tablets are available in the Netherlands (see appendix 8.2), but they claim to “help break down different types of food”, and make no mention of reducing flatulence.
- iv. As intestinal gas can result from multiple sources, taking Beano may not be effective for every consumer. Unless it is established per user that consuming legumes causes an unacceptable amount of intestinal gas for them, it might not be reasonable to buy and consume Beano regularly.

Pre-digested, flatulence free legume flours:

- i. As seen in relevant literature, it is possible to reduce the amount of RFOs in legumes by 100% if the pre-digestion method is used. If it is established that legumes are, in fact, causing the consumer’s flatulence, then eating pre-digested legume flour will certainly solve this issue.

- ii. There will be increased costs for labor, energy and materials with this method. Purified enzyme used for this experiment cost €168,00 for 2000 units. It is possible that in industrial quantities, α -galactosidase can be procured for a lesser cost, but that could not be verified for this experiment. Regardless, the process would require machinery to be set up and cleaned in place regularly. This would also add to energy costs. Therefore, if produced, the end product is likely to be significantly more expensive than the traditional legume which is attractive to customers, among other reasons, for the low price (Desrochers and Brauer, 2001).

5.4. Non-enzymatic methods of reducing RFOs in legumes

In order to evaluate a method to be made commercial, it is also relevant to consider non-enzymatic methods of reducing RFOs in legumes, and to a lesser extent, reducing intestinal gas altogether. This comprehensive approach allows us to find the most practical method overall.

Non-enzymatic methods of reducing/removing RFOs can be divided into processing strategies, and microbiological strategies (Kannan et al., 2018). Post harvest processing strategies are also used in households in order to improve the experience of legume consumption (Song and Chang, 2006; Mansour and Khalil, 1998). A basic Google search using the terms “degassing beans” or “no gas beans” yields several results, offering methods such as using baking soda, and soaking overnight (Appendix 8.3).

Physical processing methods such as boiling, autoclaving, germinating or soaking do reduce the amount of RFOs in the legumes, but to a limited extent. In pinto beans, soaking for 16 hours at room temperature reduced RFOs by 9.8%, while boiling for 90 minutes reduced them by 52.4%, and autoclaving for 30 minutes reduced them by 57.6% (Song and Chang, 2006). While it is established that reducing the amount of RFOs in a food sample does correspond to a lesser amount of intestinal gas buildup (Di Stefano, 2000),

it is still unclear how much reduction must take place in order to eliminate symptoms altogether. It is reasonable, however, to conclude that zero RFOs would correspond to zero symptoms (in an otherwise healthy subject).

Therefore, for the consumer who is not satisfied with a reduction in symptoms, but requires a complete removal of intestinal gas, other methods which result in 100% reduction of RFOs must be explored.

For example, Valentine et al., (2017) silenced the raffinose synthase gene in soybeans, successfully showing significant reduction in amount of raffinose (99%) and stachyose (98%) in the soybeans harvested. This level of reduction is on par with the enzymatic method. However, the restrictions that bind the commercialization of this method are the same as those for other genetically modified organisms. To start with, government organisations like the US Department of Agriculture or the Food and Agriculture Organization in Europe may not allow the mass production and farming of these crops. If they do become commonplace, then they may not be economically viable, as in the case of the Flavr Savr tomato in 1999 (Breuning and Lyons, 2000).

To avoid the label of 'genetically modified', legumes can also be digested bacterially by *Lactobacillus fermentum* or *Lactobacillus plantarum* (Duszkiewicz-Reinhard et al., 1994). However, this method is limited by the yield: after the longest incubation period (72 hours), the reduction in stachyose was only seen to be 27% and 43% in pinto bean and field pea, respectively.

There is also medication that can be used to control the gas itself, as opposed to reducing the RFOs. These treatments include taking simethicone, activated charcoal and anti-microbial drugs. Rifaximin, a non-absorbable antibiotic, did show significant reduction in breath H_2 and symptoms of intestinal gas. The other methods (simethicone, activated charcoal) did not show significant reduction in symptoms. (Di Stefano et al., 2000).

5.5. Flatulence-free legumes in an environmental context

As seen in the introduction of this work, many food related industries are environmentally unsustainable. This is especially true for the meat industry (Steinfeld, 2006). As established in section 1.3, legumes are an excellent substitute for meat, in terms of nutrition. In order to reduce meat consumption, it would be beneficial for consumers to switch from eating meat to eating beans.

The likelihood of consumers in 2020 choosing legumes over meat is quite high, considering the dietary shifts that are taking place globally in response to climate and water crises (Kim et al., 2019). Many consumers are held back from eating legumes in large quantities due to several reasons, including lack of familiarity and knowledge of preparation. Another common reason given is flatulence. Consumers have stated that they would consume more legumes if not for the issue of intestinal gas and flatulence (Desrochers and Brauer, 2001).

Therefore, there is cause to believe that the commercialization of a process that reduces RFOs in legumes would be beneficial to the global push for consumers to eat less meat and more legumes. If the demand for meat is reduced due to this, then there may be fewer meat-animals (e.g., poultry, and more specifically pigs and cattle) that are bred in order to provide for the meat industry. A reduction in the volume of such animals being farmed would lead to a reduction in greenhouse gas emissions from farms, while also helping to reduce the nitrogen load on the soil (Steinfeld, 2006).

On the other hand, an increase in the amount of legumes farmed also appears to have multiple benefits in the context of agricultural sustainability. Legume crops have large-scale usability due to their potential to be used as human food and cattle feed. In this context, it is beneficial to note that compared to other crops, they release 5 - 7 times less greenhouse gas emissions (Stagnari et al., 2017). They also help to save on fossil energy inputs thanks to N fertilizer reduction; legume crops have shown to fix soil nitrogen effectively thanks to strains of *Rhizobium* from the root nodules of these crops (Surange et al., 1997).

Due to the nitrogen fixing abilities of legume crops, lower amounts of nitrogen fertilizer needs to be applied to other crops if they are present in a rotation with legumes. Legume crops need no nitrogen fertilizer at all. This leads to benefits such as reduced energy use, less global warming potential, less acidification, and less ozone formation, as demonstrated in four locations in Europe (Nemecek et al., 2017). One of the regions examined in this study, Saxony-Anhalt in Germany, worked with a crop rotation involving rapeseed oil, winter wheat and winter Barley. This corresponds to crop rotations seen in the Netherlands (SenterNovem, 2005). The success of introducing legumes into the crop rotation in Saxony-Anhalt implies similar successes in the Netherlands. This could contribute to aiding the current nitrogen crisis in the Netherlands.

With these factors considered, there appear to be several benefits to supplementing or replacing meat, especially beef, with legumes. The results of this study show that one of the chief concerns regarding beans — i.e., flatulence — can be resolved using several methods.

6. Outlook

Time and resources were restricted for this project, which can account for errors that prevent the results from being as accurate as possible. To optimize this experiment in the future, the following steps may be taken:

- i. More accurate data modelling:
 - a. A greater quantity and variety of samples should be tested. This study focused on commonly-eaten legumes in the Netherlands. However, the data would be more accurately modeled if a greater variety of samples were tested. In this work, samples were procured from a local supermarket. However, in order to optimize the experiment, it may be useful to source the legumes from a research facility or a farm that has accurate information about the various strains. Additionally, multiple strains of the same legume should be tested in order to evaluate variance within that legume.

- b. Testing the substrates against various concentrations of the enzyme, as opposed to the binary nature of this study. This study used enzyme solutions of one concentration (60 U/mL). If concentrations ranging from 20 U/mL until 150 U/mL had been tested against the substrates, a Michaelis-Menten model could be established.
- c. Amount of RFO in the substrates ranged between 20 mgg⁻¹ and 100 mgg⁻¹ (observed; theoretical values ranged between 20 mgg⁻¹ and 150 mgg⁻¹), however these are not evenly distributed. In the future, more substrates should be sampled. Ideally these samples have a good distribution of quantity of RFOs. This will also allow better data modelling.
- d. During digestion with Beano, the contents of the simulated stomach environment can be sampled at regular intervals in order to establish the amount of reduction of RFOs over time.
- e. Digestion with Beano would also provide a more accurate model if the conditions of the human stomach were better replicated. For instance, with a dynamic gastric model, which would simulate the flows of the stomach better.

ii. Minimizing equipment error:

The test kit used provided a simple way to accurately quantify the amounts of RFO present in the sample. However, this experiment did not yield 100% removal of RFOs as seen in the studies performed by Song & Chang (2006). The process of digestion was not the same; instead it was a generalized process based on their study.

Not considering the differences in processing, some of the error may come from the equipment used. Instead of a water bath with manual agitation, as done in this experiment, a rotary shaker should be used in order to ensure the constant, even agitation of all the samples simultaneously.

Furthermore, the spectrophotometer used may have contributed to the error, as it was an older model.

- iii. Broadening the scope of the study to include non-enzymatic methods of RFO reduction/removal. There is a lot of variance within the enzymatic digestion method — e.g., the time of digestion of RFOs (before or after ingestion), or the source of the α galactosidase (*Aspergillus niger*, *Cladosporium cladosporoides*, *Aspergillus oryzae*, green coffee beans, etc.).

This allows in-depth studies to be conducted to find the optimal enzymatic method, especially with regard to the commercialization of legumes that do not cause flatulence. However, other methods must also be taken into account in order to have a broader perspective. For instance, the RFOs present in the legumes may also be digested by bacterial fermentation before consumption, as examined by Duszkievicz-Reinhard et al (1994).

The works of Song and Chang (2006) and Mansour and Khalil (1998) - among others - do compare the extent of RFO reduction between enzymatic methods and 'home cooking' methods. However, this can be expanded upon by including bacterial pre-digestion as well.

Furthermore, in context of commercializing flatulence-free legumes, it is important to evaluate process costs for each of these methods as well. Comparing several categories of methods can also include an economic analysis to check which process is most likely to make it to the market. Alongside establishing economic viability, logistical factors must also be considered; e.g., labelling of the product, approval from governing bodies, etc. Additionally, a social study may be performed in order to predict consumer responses to the products.

The future of this study would see these evaluations being conducted with a software such as SuperPro Designer; this would allow the researcher to have a

firm grasp on the logistical factors before beginning a collaboration with, for instance, a company that deals with food and agricultural technology.

- iv. Comparing data gathered from human trials with quantitative RFO reduction. The intention of most studies to reduce/remove RFOs before they reach the gut, is to address the issue of flatulence caused by legumes (and other α -galactoside containing foods). Thus, furthering this study would require testing in humans.

Studies such as “Does Beano Prevent Gas” (Ganiats et al., 1992) and the US patent for Beano (Kligerman, 1999) tested the product on human subjects, relying on their self-reported symptoms and breath H_2 as the basis for evaluating the effectiveness of Beano. These models can be combined with quantitative data regarding the amount of RFOs.

Human trials could include the establishment of a threshold; i.e., what quantity of RFOs would a subject have to consume in order to feel flatulent symptoms? This can be done if the researcher is aware of the quantity of RFOs present in the food being given to the subjects. It might not be necessary to remove 100% of the RFOs in order to relieve flatulent symptoms - this could be established by combining this model with a human trial.

7. References

1. Albala, K., 2007. Beans: A history. Berg Publishers, Oxford, 256.
2. Anilkumar, R. R., Edison, L. K., Pradeep, N. S., 2017. Patra, J. K., Das, G., Vishnuprasad, C. N. (Editors): Microbial Biotechnology: Volume 1. Applications in Agriculture and Environment. Springer Nature, Singapore. 135 - 162.
3. Berggen, T., Marchand, R., Barrios, A., Mitchener, C., Jones, A., Scherr, R. E., Zidenberg-Cherr, S., 2018. Nutrition and Health Info Sheet: The Paleo Diet For Health Professionals. Center for Nutrition in Schools Department of Nutrition University of California, Davis, 4.

4. Bruening G, Lyons J., 2000. The case of the FLAVR SAVR tomato. *California Agriculture* 54(4), 6-7.
5. Camilleri, M., Colemont, L. J., Phillips, S. F., Brown, M. L., Thomforde, G. M., Champan, N., Zinsmeister, A. R., 1989. Human gastric emptying and colonic filling of solids characterized by a new method. *The American Journal of Physiology* 257(2 Pt 1):G284-90.
6. CBI Ministry of Foreign Affairs, 2019. Exporting canned beans and pulses to Europe. [Online] Available: <https://www.cbi.eu/market-information/processed-fruit-vegetables-edible-nuts/canned-beans/>
7. Chiles, R. M. and Fitzgerald, A. J., 2017. Why is meat so important in Western history and culture? A genealogical critique of biophysical and political-economic explanations. *Agricultural and Human Values* 35(1), 1 - 17.
8. Den Hartog, L.A. and Sijtsma, R., 2011. The future of animal feeding: towards sustainable precision livestock farming. *Banff Pork Seminar Proceedings. Advances in Pork Production* 22, 1-16.
9. Desrochers, N., and Brauer, P. M., 2001. Legume promotion in counselling: an e-mail survey of dietitians. *Canadian Journal Of Dietetic Practice And Research* 62 (4), 193-198,
10. Dey, P. M., 1980. Biochemistry of α -D-Galactosidic Linkages in the Plant Kingdom. *Advances in Carbohydrate Chemistry and Biochemistry* Volume 37, 283–372.
11. Di Stefano, M., Strocchi, A., Malservisi, S., Veneto, G., Ferrieri, A., & Corazza, G. R., 2000. Non-absorbable antibiotics for managing intestinal gas production and gas-related symptoms. *Alimentary Pharmacology and Therapeutics*, 14(8), 1001–1008.
12. Di Stefano, M., Miceli, E., Gotti, S., Missanelli, A., Mazzocchi, S., Corazza, G. R., 2007. The Effect of Oral α -Galactosidase on Intestinal Gas Production and Gas-Related Symptoms. *Digestive Diseases and Sciences* 52, 78 - 83.
13. Duszkievicz-reinhard, W., Gujska, E. and Khan, K., 1994. Reduction of Stachyose in Legume Flours by Lactic Acid Bacteria. *Journal of Food Science*, 59: 115-117

14. FoodData Central (FDC), 2019. Beef, grass-fed, strip steaks, lean only, raw. U.S. Department of Agriculture, Agricultural Research Service. Retrieved 06/01/2020 from <https://fdc.nal.usda.gov/fdc-app.html#/food-details/169429/nutrients>
15. FoodData Central (FDC), 2019. Chicken, broilers or fryers, breast, meat and skin, raw. U.S. Department of Agriculture, Agricultural Research Service. Retrieved 06/01/2020 from <https://fdc.nal.usda.gov/fdc-app.html#/food-details/171474/nutrients>
16. FoodData Central (FDC), 2019. Chickpeas (garbanzo beans, bengal gram), mature seeds, raw. U.S. Department of Agriculture, Agricultural Research Service. Retrieved 06/01/2020 from <https://fdc.nal.usda.gov/fdc-app.html#/food-details/173756/nutrients>
17. FoodData Central (FDC), 2019. [HISTORICAL RECORD]: Beans, Dry, Dark Red Kidney (0% moisture). U.S. Department of Agriculture, Agricultural Research Service. Retrieved 06/01/2020 from <https://fdc.nal.usda.gov/fdc-app.html#/food-details/335245/nutrients>
18. FoodData Central (FDC), 2019. [HISTORICAL RECORD]: Beans, Dry, Pinto (0% moisture). U.S. Department of Agriculture, Agricultural Research Service. Retrieved 06/01/2020 from <https://fdc.nal.usda.gov/fdc-app.html#/food-details/335695/nutrients>
19. FoodData Central (FDC), 2019. Lamb, New Zealand, imported, loin chop, separable lean only, raw. U.S. Department of Agriculture, Agricultural Research Service. Retrieved 06/01/2020 from <https://fdc.nal.usda.gov/fdc-app.html#/food-details/173810/nutrients>
20. FoodData Central (FDC), 2019. Mung beans, mature seeds, raw. U.S. Department of Agriculture, Agricultural Research Service. Retrieved 06/01/2020 from <https://fdc.nal.usda.gov/fdc-app.html#/food-details/174256/nutrients>
21. FoodData Central (FDC), 2019. Peas, green, split, mature seeds, raw. U.S. Department of Agriculture, Agricultural Research Service. Retrieved 06/01/2020 from <https://fdc.nal.usda.gov/fdc-app.html#/food-details/172428/nutrients>
22. FoodData Central (FDC), 2019. Pork, Leg Cap Steak, boneless, separable lean and fat, raw. U.S. Department of Agriculture, Agricultural Research Service.

- Retrieved 06/01/2020 from <https://fdc.nal.usda.gov/fdc-app.html#/food-details/169186/nutrients>
23. FoodData Central (FDC), 2019. RED LENTILS. U.S. Department of Agriculture, Agricultural Research Service. Retrieved 06/01/2020 from <https://fdc.nal.usda.gov/fdc-app.html#/food-details/551553/nutrients>
 24. FoodData Central (FDC), 2019. Soybeans, mature seeds, dry roasted. U.S. Department of Agriculture, Agricultural Research Service. Retrieved 06/01/2020 from <https://fdc.nal.usda.gov/fdc-app.html#/food-details/172441/nutrients>
 25. Ganiats, T. G., Norcross, W. A., Halverson, A. L., Burford, P. A., Palinkas, L. A., 1994. Does Beano Prevent Gas?: a double-blind crossover study of oral alpha-galactosidase to treat dietary oligosaccharide intolerance. *Journal of Family Practice* 5 (39), 441 - 445.
 26. Guce, A. I., Clark, N. E., Salgado, E. N., Ivanen, D. R., Kulminskaya, A. A., Burmer, H., Garman, S. C., 2009. Catalytic Mechanism of Human α -Galactosidase. *The Journal of Biological Chemistry* 285, 3625 - 3632.
 27. Han, I. H. and Baik, B. 2006. Oligosaccharide Content and Composition of Legumes and Their Reduction by Soaking, Cooking, Ultrasound, and High Hydrostatic Pressure. *Cereal Chemistry Journal*, 83 (4), 428-433.
 28. Jung, M., and Lee, S., 1998. Stability of acetal and non acetal-type analogs of artemisinin in simulated stomach acid. *Bioorganic & Medicinal Chemistry Letters*, 8(9), 1003–1006.
 29. Hoffman, J. R., and Falvo, M. J., 2005. Protein – Which is Best? *Journal of Sport Science and Medicine* 3(3), 118 - 130.
 30. Kannan, U., Sharma, R., Gangola, M. P., Chibbar, R. M., 2018. Improving Grain Quality in Pulses: Strategies to Reduce Raffinose Family Oligosaccharides in Seeds. *Journal of Crop Breeding and Genetics* 4 (1), 70 - 88.
 31. Kantor, L. S., 1998. A Dietary Assessment of the U.S. Food Supply: Comparing Per Capita Food Consumption with Food Guide Pyramid Serving Recommendations. Economic Research Service, U. S. Department of Agriculture. 1 - 55.

32. Katrolia, P., Rajashekhara, E., Yan, Q., Jiang, Z., 2014. Biotechnological potential of microbial α -galactosidases. *Critical Reviews in Biotechnology* 34 (4), 307 - 317.
33. Khan, M. N., and Mohammed, F., 2014. Eutrophication: Challenges and Solutions. Ansari, A. A., Gill, S. S., (Editors) *Eutrophication: Causes, Consequences and Control*. Springer, Dordrecht, 1 - 15.
34. Kleintop, A. E., Echeverria, D., Brick, L. A., Thompson, H. J., Brick, M. A., 2013. Adaptation of the AOAC 2011.25 Integrated Total Dietary Fiber Assay to Determine the Dietary Fiber and Oligosaccharide Content of Dry Edible Beans. *Journal of Agricultural and Food Chemistry* 61 (40), 9719 - 9726.
35. Kligerman, A. E., 1999. Composition and method for reducing gastro-intestinal distress due to alpha-D-galactoside-linked/containing sugars. U. S. Patent number 5,989,544. Washington, D.C: U.S. Patent and Trademark Office.
36. Kotiguda, Girigowda, Kapnoor, S. S., Kulkarni, D., and Mulimani, V. H., 2006. Degradation of Raffinose Oligosaccharides in Soymilk by Immobilized α -Galactosidase of *Aspergillus oryzae*. *Journal of Microbiology and Biotechnology* 17 (9), 1430 - 1436.
37. Larijani, B., Esfahani, M. M., Moghimi, M., Ardakani, M. R. S., Keshavarz, M., Kordafshari, G., Nazem, E., Ranjbar, S. H., Kenari, H. M., Zargarani, A., 2016. Prevention and Treatment of Flatulence From a Traditional Persian Medicine Perspective. *Iran Red Crescent Medical Journal* 18(4):e23664.
38. Makkar, H. P. S., and Vercoe, P. E., 2007. Introduction. Makkar, H. P. S., and Vercoe, P. E. (Editors) *Measuring Methane Production from Ruminants*. Springer, Dordrecht, ix - xi.
39. Manheimer, E. W., van Zuuren, E. J., Fedorowicz, Z., and Pijl, H., 2015. Paleolithic nutrition for metabolic syndrome: systematic review and meta-analysis. *The American journal of clinical nutrition*, 102(4), 922–932.
40. Mansour, E. H., and Khalil, A. H., 1998. Reduction of Raffinose Oligosaccharides in Chickpea (*Cicer arietinum*) Flour by Crude Extracellular Fungal α -Galactosidase. *Journal of the Science of Food and Agriculture* 78, 175 - 181.

41. Mao, B., Tang, H., Gu, J., Li, D., Cui, S., Zhao, J., Zhanga, H., Chen, W., 2018. In vitro fermentation of raffinose by the human gut bacteria. *Food and Function* 9, 5824 - 5831.
42. Martínez-Villaluenga, C., Frias, J., & Vidal-Valverde, C., 2006. Alpha-Galactosides: Antinutritional Factors or Functional Ingredients? *Critical Reviews in Food Science and Nutrition*, 48(4), 301–316.
43. McCleary, B.V., Charnock, S.J., Rossiter, P.C., O'Shea, M.F., Power, A.M., Lloyd, R.M., 2006. Measurement of carbohydrates in grain, feed and food. *Journal of the Science of Food and Agriculture* 86, 1648–1661.
44. McPhee, K. E., Zemetra, R. S., Brown, J., Myers, J. R., 2002. Genetic Analysis of the Raffinose Family Oligosaccharides in Common Bean. *Journal of the American Society for Horticultural Science* 127 (3), 376 - 382.
45. Megazyme International, 2018. Raffinose/Sucrose/Glucose Assay Procedure. [Online] Available: https://www.megazyme.com/documents/Booklet/K-RAFGL_DATA.pdf
46. National Center for Biotechnology Information (NCBI). PubChem Database. Ajugose, CID=441421, <https://pubchem.ncbi.nlm.nih.gov/compound/Ajugose> (accessed on Jan. 31, 2020)
47. National Center for Biotechnology Information (NCBI). PubChem Database. Ciceritol, CID=10142653, <https://pubchem.ncbi.nlm.nih.gov/compound/Ciceritol> (accessed on Jan. 28, 2020)
48. National Center for Biotechnology Information (NCBI). PubChem Database. D-Glucose, CID=5793, <https://pubchem.ncbi.nlm.nih.gov/compound/D-Glucose> (accessed on Jan. 21, 2020)
49. National Center for Biotechnology Information (NCBI). PubChem Database. Raffinose, CID=439242, <https://pubchem.ncbi.nlm.nih.gov/compound/Raffinose> (accessed on Jan. 28, 2020)
50. National Center for Biotechnology Information (NCBI). PubChem Database. Stachyose, CID=439531, <https://pubchem.ncbi.nlm.nih.gov/compound/Stachyose> (accessed on Jan. 21, 2020)

51. National Center for Biotechnology Information (NCBI). PubChem Database. Sucrose, CID=5988, <https://pubchem.ncbi.nlm.nih.gov/compound/Sucrose> (accessed on Jan. 21, 2020)
52. National Center for Biotechnology Information (NCBI). PubChem Database. Verbascose, CID=441434, <https://pubchem.ncbi.nlm.nih.gov/compound/Verbascose> (accessed on Jan. 28, 2020)
53. Nemecek, T., von Richthofen, J., Dubois, G., Casta, P., Charles, R., Pahlf, H. 2017. Environmental impacts of introducing grain legumes into European crop rotations. *European Journal of Agronomy* 28, 380 - 393.
54. Newton, B. E., 1975. "The Nitrosation of Foods." PhD diss., University of Surrey, 1975. ProQuest (doi:10.1042/bj1300082pa)
55. Paleo leap LLC, (2019). What's wrong with beans and legumes? [Weblog]. Retrieved 28 December 2019, from <https://paleoleap.com/beans-and-legumes/>
56. Patil, A. G. G., Kote, N. V., Mulimani, V., 2009. Enzymatic removal of flatulence-inducing sugars in chickpea milk using free and polyvinyl alcohol immobilized α -galactosidase from *Aspergillus oryzae*. *Journal of Industrial Microbiology and Biotechnology* 36, 29 - 33.
57. Purchas, R. W., Busboom, J. R., & Wilkinson, B. H. P., 2006. Changes in the forms of iron and in concentrations of taurine, carnosine, coenzyme Q10, and creatine in beef longissimus muscle with cooking and simulated stomach and duodenal digestion. *Meat Science*, 74(3), 443–449.
58. Quemener, B., & Brillouet, J.-M., 1983. Ciceritol, a pinitol digalactoside form seeds of chickpea, lentil and white lupin. *Phytochemistry*, 22(8), 1745–1751.
59. SenterNovem, 2005. The road to pure plant oil? The technical, environment-hygienic and cost-related aspects of pure plant oil as a transport fuel. Dutch Ministry for Spatial Planning, Housing and the Environment.
60. Smýkal, P., Vernoud, V., Blair, M. W., Soukup, A., Thompson, R., 2014. The role of the testa during development and in establishment of dormancy of the legume seed. *Frontiers in Plant Science* 5, 1 - 351.

61. Song, D. and Chang, S. K. C., 2006. Enzymatic Degradation of Oligosaccharides in Pinto Bean Flour. *Journal of Agriculture and Food Chemistry* 54, 1296 - 1301.
62. Stagnari, F., Maggio, A., Galieni, A., Pisante, M., 2017. Multiple benefits of legumes for agriculture sustainability: an overview. *Chemical and Biological Technologies in Agriculture*, 4(2) .
63. Statistics Netherlands (CBS), 2018. The Sustainable Development Goals: the situation for the Netherlands. Statistics Netherlands, The Hague, 104.
64. Steinfeld, H., 2006. *Livestock's Long Shadow: environmental issues and options*. Food and Agriculture Organization of the United Nations, Rome, 416.
65. Suarez, F. L., Springfield, J., Furne, J. K., Lohrmann, T. T., Kerr, P. S., Levitt, M. D., 1999. Gas production in humans ingesting a soybean flour derived from beans naturally low in oligosaccharides. *The American Journal of Clinical Nutrition* 69, 135 - 139.
66. Surange, S., Wollum II, A. G., Kumar, N., Nautiyal, C. S., 1997. Characterization of *Rhizobium* from root nodules of leguminous trees growing in alkaline soils. *Canadian Journal of Microbiology* 43 (9), 891 - 894.
67. Taraday, J (2018). 5 Reasons You Shouldn't Avoid Eating Beans. [Weblog]. Retrieved 28 December 2019, from <https://breakingmuscle.com/healthy-eating/5-reasons-you-shouldn-t-avoid-eating-beans>
68. Tester, R. F., Karkalas, J, 2003. CARBOHYDRATES | Classification and Properties. Caballero, B. (Editor-in-Chief): *Encyclopedia of Food Sciences and Nutrition*. Academic Press, Cambridge, 862 - 875.
69. Trujillo Toledo, L. E., Martinez García, D., Pérez Cruz, E., Rivera Intriago, L. M., Pérez, J. N., and Pais Chanfrau, J. M. (2019). Fructosyltransferases and Invertases: Useful Enzymes in the Food and Feed Industries. *Enzymes in Food Biotechnology*, 451–469.
70. Tweede Kamer der Staten-Generaal, 2019. Regels voor de aanpak van de stikstofproblematiek in relatie tot natuur (Spoedwet aanpak stikstof). Tweede Kamer, 's-Gravenhage, 35 347, nr. 2.
71. Valentine, M. F., De Tar, J. R., Muruganantham, M., Firman, J. D., Zhang, Z. J., 2017. Silencing of Soybean Raffinose Synthase Gene Reduced Raffinose Family

Oligosaccharides and Increased True Metabolizable Energy of Poultry Feed. Frontiers in Plant Science, 8.

72. Vidal-Valverde, C., Frías, J., and Valverde, S., 1993. Changes in the carbohydrate composition of legumes after soaking and cooking. Journal of the American Dietetic Association, 93(5), 547–550.
73. Winham, D. M. and Hutchins, A. M., 2011. Perceptions of flatulence from bean consumption among adults in 3 feeding studies. Nutrition Journal 10(128), 1 - 9.

8. Appendix

8.1 Specifics about Beano

Note: Images seen in this section are photographs that were taken of the package of Beano used for this experiment.

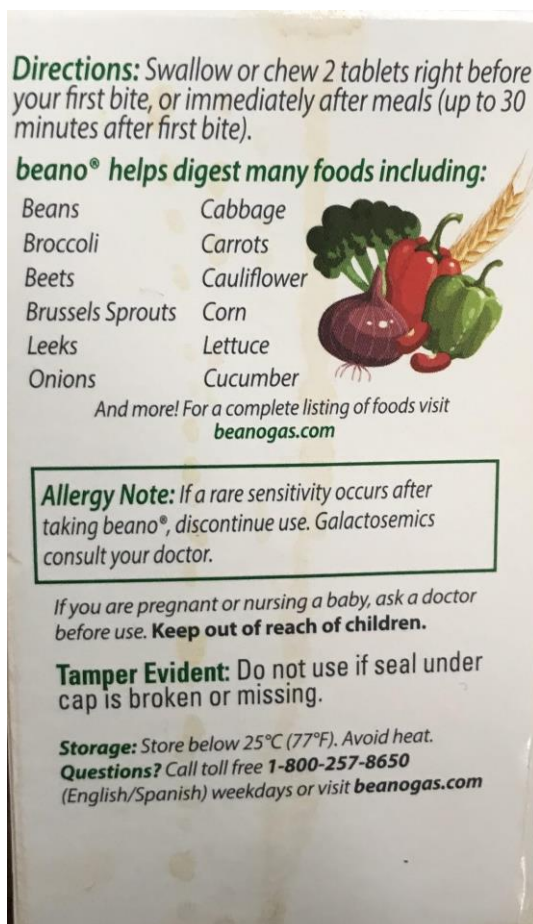


Figure 8.1.1 : Directions of use of Beano. It also indicated 'problem foods' such as broccoli, cauliflower, and cabbage (cruciferous foods containing raffinose); cucumber (containing cucurbitacin, which occurs as glycosides); as well as the general category of beans. The image also specifies that "a rare sensitivity" may occur, and that patients with galactosemia must consult their doctor before use.

beano[®]
Food Enzyme
Dietary Supplement

***Stop discomfort
before it starts!****

How beano[®] works: beano[®] contains a natural enzyme that helps digest the complex carbohydrates found in many foods, so they don't cause gas, bloating and discomfort*.

***THESE STATEMENTS HAVE NOT BEEN EVALUATED BY THE FOOD AND DRUG ADMINISTRATION. THIS PRODUCT IS NOT INTENDED TO DIAGNOSE, TREAT, CURE OR PREVENT ANY DISEASE.**

Figure 8.1.2: “How beano works”. explaining pictorially the digestion of RFOs in the stomach before reaching the colon. Image also specifies that the statements have not been verified by the Food & Drug Administration; indicating that Beano is not a pharmaceutical product.

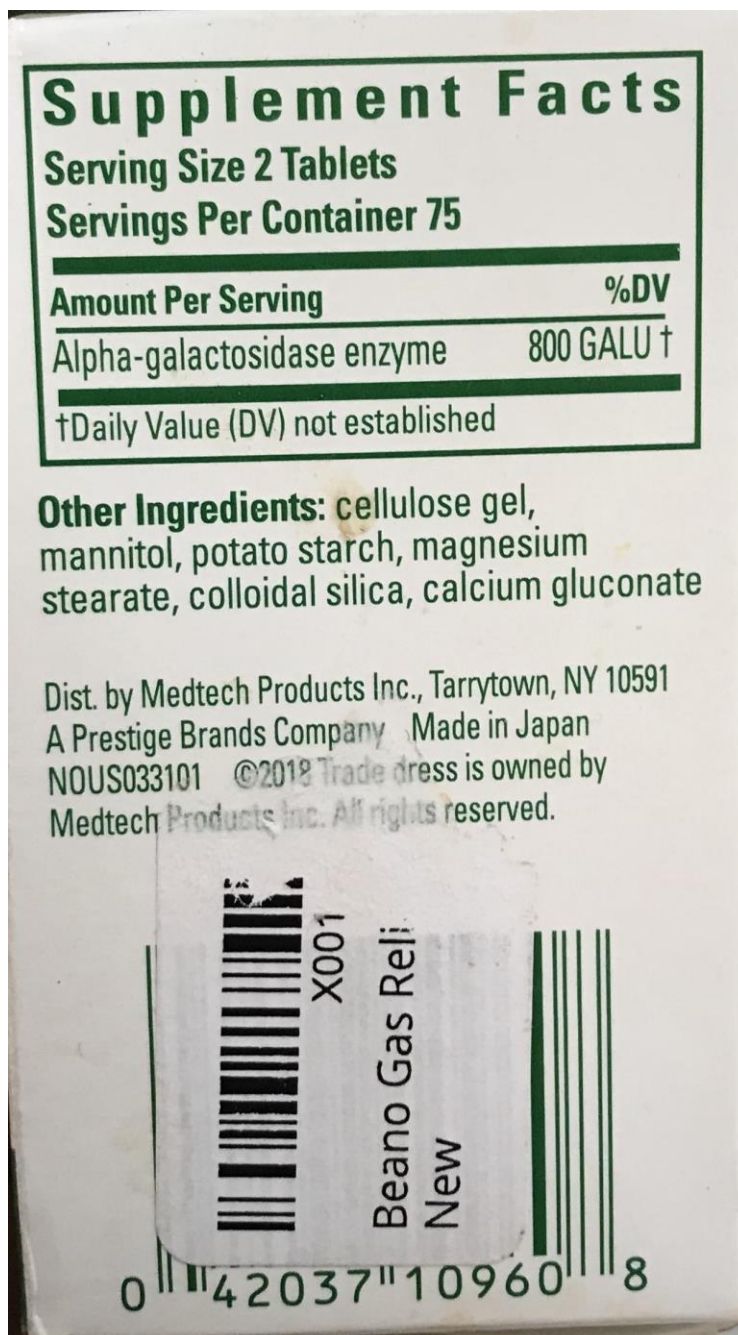


Figure 8.1.3: Supplement facts. Serving size included. It indicates that 2 tablets contain 800 Units of the enzyme. Image also shows other ingredients, used as stabilizers for the enzyme.

8.2 Specifics regarding other enzyme supplements

Holland & Barrett MultiDigestive Enzyme 90 Tablets £4.24 was £8.49

★★★★★ 4.4 (57)



Size:
90 Tablets

16 rewards for life points
SKU: 030667

1/2 Price

was £8.49
now **£4.24**
£0.05/Tablet
✓ In Stock

Quantity:

☒ Deliver one time only

☐ Subscribe & Save [Info](#)
On future orders get the best price from the last 90 days plus up to 15% off & free delivery. Pause or cancel any time.

NEW Get every 5th order free

Deliver every:

Add to Basket

♥ Add to favourites

Figure 8.2.1: Multi-digestive enzyme tables are available online and in brick-and-mortar stores in the Netherlands, and are relatively inexpensive. For comparison, the Beano used in this experiment cost £30,00.

Description

- Multi-digestive Enzyme supplement with added benefits of Betaine HCl
- Suitable for vegetarians and vegans
- Quality Assured; high quality and rigorously tested

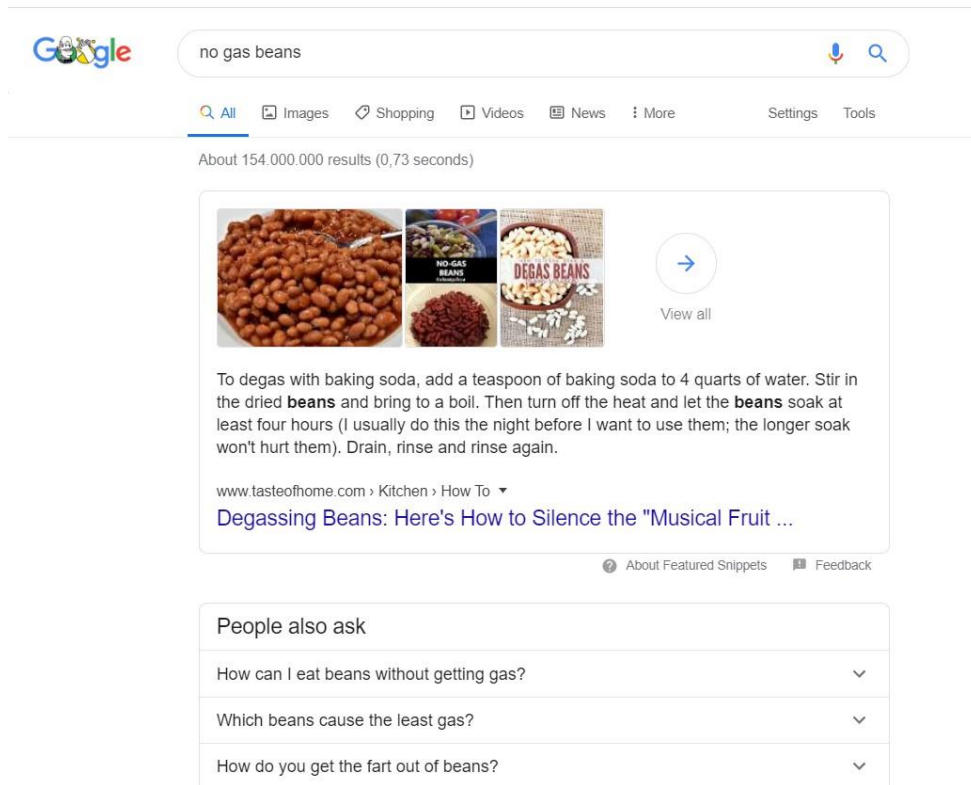
These tablets contain protein, fat and carbohydrate digesting enzymes with additional Betaine HCl. Best taken just before or directly after eating, the enzymes help to break down different food types and the betaine HCl.

Figure 8.2.2: Description of the multi enzyme tablets.

Nutritional Information	
MultiDigestive Enzyme each tablet contains:	
Amylase	11mg
Betaine Hydrochloride	100mg
Bromelain	22mg
Lipase	8.25mg
Oil of Peppermint	1mg
Papain	8mg
Show Less	

Figure 8.2.3: Information about enzymes in the Multi Enzyme tablets.

8.3. “No gas beans”



The image is a screenshot of a Google search results page for the query "no gas beans". At the top, the Google logo is on the left, and the search bar contains "no gas beans" with a microphone icon and a search button. Below the search bar are tabs for "All", "Images", "Shopping", "Videos", "News", "More", "Settings", and "Tools". The search results show "About 154,000,000 results (0,73 seconds)". A featured snippet is displayed, containing three images: a bowl of cooked beans, a bowl of beans with a label "NO-GAS BEANS", and a bowl of beans with a label "DEGAS BEANS". To the right of the images is a "View all" button with a right arrow. Below the images, the text reads: "To degas with baking soda, add a teaspoon of baking soda to 4 quarts of water. Stir in the dried **beans** and bring to a boil. Then turn off the heat and let the **beans** soak at least four hours (I usually do this the night before I want to use them; the longer soak won't hurt them). Drain, rinse and rinse again." Below this text is the source "www.tasteofhome.com > Kitchen > How To" and a link titled "Degassing Beans: Here's How to Silence the 'Musical Fruit ...". At the bottom of the featured snippet are links for "About Featured Snippets" and "Feedback". Below the featured snippet is a "People also ask" section with three questions: "How can I eat beans without getting gas?", "Which beans cause the least gas?", and "How do you get the fart out of beans?". Each question has a downward arrow icon.

Google

no gas beans

All Images Shopping Videos News More Settings Tools

About 154,000,000 results (0,73 seconds)



[View all](#)

To degas with baking soda, add a teaspoon of baking soda to 4 quarts of water. Stir in the dried **beans** and bring to a boil. Then turn off the heat and let the **beans** soak at least four hours (I usually do this the night before I want to use them; the longer soak won't hurt them). Drain, rinse and rinse again.

www.tasteofhome.com > Kitchen > How To

[Degassing Beans: Here's How to Silence the "Musical Fruit ...](#)

[About Featured Snippets](#) [Feedback](#)

People also ask

How can I eat beans without getting gas? [▼](#)

Which beans cause the least gas? [▼](#)

How do you get the fart out of beans? [▼](#)

Figure 8.3.1: Blogs recommending soaking with baking soda in an effort to ‘make beans less gassy’

theboatgalley.com › no-gas-beans ▼

No-Gas Beans | The Boat Galley

Oct 19, 2011 - Use about three times as much water as **beans**, or guesstimate and cover them by at least one inch, preferably two. Cook them until tender . . . **not** ...

health.clevelandclinic.org › the-musical-fruit-what-you-should-know-... ▼

What You Should Know About Beans and the (Embarrassing ...

Sep 20, 2018 - Todorov points out, however, that 6 to 12 percent of the people saw **no** decrease in **gas** with any **bean**. "People vary in their response to different ...

dontwastethecrumbs.com › Meal Planning › Kitchen How-To's

How to Soak, De-Gas and Cook Beans from Scratch



Jul 2, 2019 - ★★★★★ Rating: 4.9 - 15 reviews

I should note that **not** all brands of **beans** are created equal. The lower-priced brands often include more ...

www.popsugar.com › fitness › How-Eat-Beans-Have-Gas-44527086 ▼

How to Eat Beans and Not Have Gas | POPSUGAR Fitness

Dec 8, 2018 - Soak dry **beans** in water for at least eight hours, but the longer the better; try overnight. Discard the water, then add new water and cook for 30 to ...

nutritionovereasy.com › 2015/10 › six-tips-for-reducing-gas-from-be... ▼

Six tips for reducing gas from beans | Nutrition Over Easy

Oct 7, 2015 - **Not** the sexiest post title I've ever written, that's for sure!) Dried **beans** and ...

Figure 8.3.2: Various methods of home-processing to reduce gas from beans are available after a Google search.

9. Statutory declaration

Declaration:

I hereby declare that I wrote the present dissertation with the topic “Evaluation of two methods to reduce legume-related flatulence through enzymatic digestion of flatulence factors” independently and used no other aids than those cited. In each individual case, I have clearly identified the source of the passages that are taken word for word, or paraphrased from other works.

I also hereby declare that I have carried out my scientific work according to the principles of good scientific practice.

Tilburg, 27/04/2020

Shraddha Ranganathan